# BASE24 AKDS IMPLEMENTATION GUIDE



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## 1. PREFACE

**a) Required Reading**

Prior to installing AKDS, you should review all of the following AKDS-related documents available on InfoLink:

- ***BASE24-atm Diebold 10XX/478X Device Support Manual***
- ***BASE24-atm NCR 5XXX Device Support Manual***
- ***BASE24 Release 6.0 Version 3 Implementation Guide (Automated Key Distribution System Enhancement)***
- ***BASE24 Release 6.0 Version 4 Implementation Guide (Automated Key Distribution System Enhancement)***
- ***BASE24 Release 6.0 Version 5 Implementation Guide (Diebold AKDS Capability Check and Load All Keys Enhancement, NCR AKDS Capability Check and Load All Keys Enhancement)***
- ***BASE24 Release 6.0 Version 6 Implementation Guide (NDC+ 5XXX Device Handler AKDS Certificate Support Enhancement, Shared (Wincor) 912 AKDS Signatures Enhancement, Shared (Wincor) NDC+ AKDS Signatures Enhancement)***
- ***BASE24 Release 6.0 Version 7 Implementation Guide (Shared 912 AKDS Certificate Support Enhancement, Shared NDC+ AKDS Certificates Support Enhancement)***
- ***BASE24 Release 6.0 Version 8 Implementation Guide (Diebold 912 AKDS Signature Support Enhancement, Wincor AKDS Support of NCR ATMs)***
- ***BASE24 Release 6.0 Version 9 Implementation Guide (Tidel AKDS Support Enhancement, Triton AKDS Support Enhancement)***
- ***BASE24 Release 6.0 Version 10 Implementation Guide (TSS new AKDS message format for NCR and Diebold Enhancement)***
- ***BASE24 Release 6.0 Post Version 10 Implementation Guide (NCR Enhanced AKDS)***
- ***BASE24 Transaction Security Services Processing Guide***

### b) Assumptions

It is assumed that TSS has been installed on your system prior to performing the steps outlined in this AKDS Install Guide.

### c) Solutions

There are currently ten different AKDS solutions available for customers.
Which one a customer will use is dependent on the ATM hardware and ATM software that is deployed at the customer’s site, along with the message format (NDC+ or 912) that the customer wishes to use.

The following are the current options:

**Native Diebold 10XX/478X**
Used when the ATM is a Diebold ATM with a certificate-based EPP. The ATM’s software is either Diebold’s TCS, TCSPlus or Agilis software, or is NCR’s Aptra Edge or Aptra Advance

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software running in Diebold emulation. The message format being used is 912, via the Diebold 10XX/478X Device Handler.

**Native NCR 5XXX**
Used when the ATM is an NCR ATM with a signature-based EPP. The ATM's software is either NCR's Aptra Edge or Aptra Advance software or Diebold's Agilis software running in NCR emulation. The message format being used is NDC+, via the NCR 5XXX Device Handler. The original NCR AKDS solution is now known as the Basic scheme, and an Enhanced scheme is now also offered.

**NCR 5XXX Certificate Support**
Used when the ATM is a Diebold ATM with a certificate-based EPP. The ATM's software is NCR's Aptra Edge software. The message format being used is NDC+, via the NCR 5XXX Device Handler.

**Diebold 912 Signature Support**
Used when the ATM is an NCR ATM with a signature-based EPP. The ATM's software is Diebold's Agilis 91x software. The message format being used is 912, via the Diebold 10XX/478X Device Handler

**Shared 912 Signatures Support**
Used when the ATM is an NCR or Wincor ATM with a signature-based EPP. The ATM's software conforms to the Shared 912 Signature specification created by Wincor Nixdorf. The message format being used is 912, via the Diebold 10XX/478X Device Handler.

**Shared NDC+ Signatures Support**
Used when the ATM is an NCR or Wincor ATM with a signature-based EPP. The ATM's software conforms to the Shared NDC+ Signature specification created by Wincor Nixdorf. The message format being used is NDC+, via the NCR 5XXX Device Handler.

**Shared 912 Certificate Support**
Used when the ATM is a Diebold ATM with a certificate-based EPP. The ATM's software conforms to the Shared 912 Certificate specification created by Wincor Nixdorf. The message format being used is 912, via the Diebold 10XX/478X Device Handler.

**Shared NDC+ Certificate Support**
Used when the ATM is a Diebold ATM with a certificate-based EPP. The ATM's software conforms to the Shared NDC+ Certificate specification created by Wincor Nixdorf. The message format being used is NDC+, via the NCR 5XXX Device Handler.

**Tidel Support**
Used when the ATM is a Tidel ATM with a signature-based EPP. The ATM's software is Tidel's 4.22.00 software. The message format being used is Tidel's, via the Tidel Device Handler.

**Triton Support**
Used when the ATM is a Triton ATM with a certificate-based EPP. The ATM's software is Triton's software, minimum version of 2.3. The message format being used is Triton's, via the Triton Device Handler.

## End of Section

| | | |
|---|---|---|
| | **AKDS PROCESS SETUP** | |
| **1.** | **APPLY UPDATED TSS SOFTWARE** | |
| **a)** | Apply all software updates using the documentation contained in the **TSS xx.x Documentation.zip** file, received when you took delivery of the new software. | |
| **2.** | **CONFIGURE THE AKDS PROCESS IN XPNET** | |
| **a)** | Determine which resource nodes in the BASE24 system will have an AKDS process. | |
| **b)** | Add an AKDS process to each selected resource node by entering the following commands using the Network Control Point Communications utility (NCPCOM) or the NCS screen. **Note:** Change the **bold** items to conform to your environment.<br><br>RESET PROCESS<br>SET PROCESS LIBRARY **$<VOL>.XNLIBN.**XNLIBN<br>SET PROCESS PROGRAM **$<VOL>.**TSS6TOBJ.AKDSISLO<br>SET PROCESS PPD **$TAKD**<br>SET PROCESS PRIORITY 147<br>SET PROCESS CPU **1**<br>SET PROCESS QAT **30**<br>SET PROCESS QMT **75**<br>SET PROCESS HOMETERM **$SUDO**<br>SET PROCESS SERVICE IS:AKDSIS<br>SET PROCESS STARTOPTIONS "-MDBV=CSV -AI=TSS<br>-STRT=**$<VOL>.**TSSTCNFG"<br>SET PROCESS DEFINES OFF<br>SET PROCESS REPLACEDEST OFF<br>SET PROCESS NSKSTARTSEQ ON<br>ADD PROCESS **P1A^AKDSLO01**, **UNDER SYSNAME \HP, UNDER NODE**<br>**P1A^NODE**<br><br>**Note:** The AKDS process is a special Transaction Security Services process that contains all of the components of the Transaction Security Services process plus the components required to support the BASE24-atm Automated Key Distribution System add-on product. The AKDS process uses the AKDSISLO executable object while the regular Transaction Security Services process uses the TSSISLO executable object. | |
| **c)** | Determine whether link processes are required between resource nodes.<br><br>If the AKDS process added above is in a different resource node from any of the device handler processes that will perform the AKDS commands, a link process will be required between the device handler's resource node and the AKDS process' resource node.<br><br>If necessary, enter the following commands using the Network Control Point Communications utility (NCPCOM) or the NCS screen. **Note:** Change the **bold** items to conform to your environment.<br><br>RESET LINK<br>SET LINK PPD $**B452B**<br>ADD LINK **P1A**^LINK^**P52B**, UNDER SYSNAME **\HP**, UNDER NODE **P1A^NODE** | |

| AKDS PROCESS SETUP | | |
|---|---|---|
| | RESET LINK<br>SET LINK PPD **$BB4AN**<br>ADD LINK **P52B**^LINK^**P1A**, UNDER SYSNAME **HP**, UNDER NODE **P52B^NODE** | |
| **3.** | **MODIFY THE MDSDEST FILE** | |
| | The Message Delivery Service Destination data file (MDSDEST) contains destinations external to a process. The location of the file is specified by the SIMDS_DEST_FILENAME parameter in the CONFCSV file on the TSSTDATA subvolume.<br><br>The MDSDEST contains one line for each destination. Each line contains the following information: Column 1 specifies the assign name. Column 2 specifies the PPD name. Column 3 specifies if raw text is sent to the destination (Y = raw send, N = sent with XPNET headers). | |
| **a)** | Rename the current MDSDEST data file to OMDSDEST. | |
| **b)** | Add the AKDS process if it is not already present. To do so, edit the MDSDESTS edit file and insert a line as shown in the following example.<br><br>• The first line of the example defines the destination of the SIS Timer Process.<br>• The second and third lines define Hardware Security Module destinations. The assigns in these lines must match the assign names defined on the Security Device Configuration window.<br>• The fourth line of the example defines the IS. This is used when the User Interface Server process sends a message to the IS process.<br>• The fifth line of the examples defines the AKDS process. This is used when the User Interface Server process sends a message to the AKDS process. **Add this line if it does not already exist.**<br>• The sixth line defines the TCP/IP Adaptor process used by the User Interface for User Security.<br><br>The following example uses an Atalla Security Device:<br><br>#STP, $TTIM, N<br>ATALLA1, $BOX1, Y<br>ATALLA2, $BOX2, Y<br>IS1, $TISL, N<br>**IS2, $TAKD, N**<br>UISEC, $TTCP.#UISEC1, Y<br><br>The following example uses a Thales e-Security (Racal) Security Device:<br><br>#STP, $TTIM, N<br>RACAL1, $DEV1, Y<br>RACAL2, $DEV2, Y<br>IS1, $TISL, N<br>**IS2, $TAKD, N**<br>ISEC, $TTCP.#UISEC1, Y | |

| AKDS PROCESS SETUP | | |
|---|---|---|
| **c)** | Now build the new MDSDEST data file.<br><br>➢ RUN $<VOL>.TSS6TOBJ.SIRDFLLO /IN MDSDESTS, LIB <$VOL>.XNLIBN.XNLIBN/ TSSTDATA.MDSDEST<br><br>**Note:** If a "File MDSDEST is not audited" warning message is displayed during this step, it can be ignored. | |
| **d)** | Use the FUP COPY command to display the contents of the MDSDEST file. An example is shown below.<br><br>#STP $TTIM N<br>ATALLA1 $BOX1 Y<br>ATALLA2 $BOX2 Y<br>IS1 $TISL N<br>IS2 $TAKD N<br>UISEC $TTCP.#UISEC1 Y | |
| **4.** | **CREATE THE AKDS DATA FILES** | |
| | AKDS requires the following additional data files. These files will reside in the TSS data subvolume, $<VOL>.TSSTDATA. The AKDS file creation schemas are found in the TSSFUP file on the TSSTCNFG subvolume, or in the SCPTENSC file on the TSSCNFG subvolume for TSS 08.2 and later. .<br><br>• CHNPKD Channel Public Key Security File<br>• EPNUMD EPP Serial Number File (optional)<br>• HSPKPD Host Public Key Security Primary File<br>• HSPKSD Host Public Key Security Secondary File (Diebold only)<br><br>For TSS 6.4 and later, HSPKPD and HSPKSD have been replaced by a single data file – HSPKD – Host Public Key Security File. | |
| **a)** | Because the TSSFUP file contains schemas for all TSS files, you should either extract the AKDS file schemas to a new file (AKDSFUP) or create all the TSS files in a temporary subvolume and move the new AKDS files to the TSS data subvolume. | |
| **b)** | To create the AKDS files, do the following.<br><br>➢ FUP DUP AKDSFUP, $<VOL>.TSSTDATA.*<br>➢ VOLUME $<VOL>.TSSTDATA<br>➢ FUP/IN AKDSFUP/ | |
| **5.** | **UPDATE THE MODULES.LST** | |
| | If AKDS is being added to an existing running TSS system, via an UPDATE cut of code, rather than a FULL cut of code, the following manual step needs to be taken to ensure that the Desktop will display the AKDS UIs. | |
| **a)** | Access the Modules.lst file under OSS in the /ESWeb/ESConfig directory.<br>Add the AKDS product ID to the list of modules supported – “SS200”. | |
| **End of Section** | | |

# ATM DEVICE SETUP

## 1. MODIFY THE ATM DEVICE CONFIGURATION IN XPNET

**a)** From the NCS screen or NCPCOM, increase the RECVLEN and XMITLEN attributes of the ATM device configuration to allow for the increased network message sizes required by AKDS. A minimum value of 2047 bytes is recommended.

>ALTER DEVICE <device name>, RECVLEN 2047
>ALTER DEVICE <device name>, XMITLEN 2047

**b)** For Tidel ATMs connected via a dialup connection, using the VISA protocol, change the device configuration for the ATM to use the ACISTD variant of the VISA protocol. This is to ensure that the connection to the ATM is not dropped after each step of the AKDS message sequence. See example device definition:

```
DEVICE: \<system>.<node>.<device name>
TYPE            255                     ACLENGTH            2
RECVLEN         2000                    XMITLEN             2000
APPLID                                  MACRO
PROFILE                                 ALTPROFILE
TRADDRESS       OFF                     TTRANSPARENT        OFF
DEFRSP          OFF                     TRLEN               0
SELECTION       NORMAL                  NRSSCP              OFF
INPUTFORMAT
OUTPUTFORMAT
POLLFORMAT
SELECTFORMAT
DESTROUTE
XNCONF
RPCONF
DEVICE:          \<system>.<node>.<device name>
WILDCHAR        "?"                     NUM DEST ENTRIES:  1

ENTRY           DESTINATION             OFFSET          MATCH CHARS
1               <Tidel DH process>      3               "A01"
DEVICE:         \<system>.<node>.<device name>
WILDCHAR        "?"                     NUM PROT ENTRIES:  1

ENTRY           PROTOCOL                OFFSET          MATCH CHARS
1               ACISTD                  0               "?"
```

# End of Section

| SECURITY DEVICE SETUP | | |
|---|---|---|
| **1.** | **ENABLE PKI FOR THE SECURITY DEVICES** | |
| | The AKDS process will choose only PKI-enabled security devices when processing AKDS commands. Therefore, PKI must be enabled in one or more security devices. | |
| **a)** | In the Security Device Configuration window of the ACI Desktop, enable PKI for each security device that will perform AKDS processing by putting a check mark in the Enable PKI box and saving the record. | |
| **b)** | Go to the Process Control window of the ACI Desktop to warmboot all configured IS processes.<br>• Select **IS** in the Destination field (**IS** is the service that the TSS processes register under, therefore all IS processes will be warmbooted).<br>• Select Alter in the Command field.<br>• Select Data Source in the Component field.<br>• Enter SEC_DEV_CONFIG in the Command Text field.<br>• Click on the green arrow in the upper left hand corner of the window to issue the command. | |
| **c)** | Check the event log for the successful completion of the warmboot by all configured IS processes. | |
| **2.** | **ENABLE REQUIRED SECURITY DEVICE COMMANDS (ATALLA)** | |
| **a)** | The AKDS product requires the following commands to be enabled in the Atalla security device(s). Enable them using the Define Security Policy command (Command 108) and Confirm Security Policy command (Command 109).<br><br>• Generate RSA Key Pair (Command 120)<br>• Generate AKB for Root Public Key (Command 122)<br>• Verify Public Key and Generate AKB of Public Key (Command 123)<br>• Generate Digital Signature (Command 124)<br>• Verify Digital Signature (Command 125)<br>• Generate Message Digest (Command 126)<br>• Generate ATM Master Key and Encrypt with Public Key (Command 12F)<br><br>**Note**: BASE24-eps AKDS also uses the Generate Random Number (command 93) command to generate a random nonce when a new master key is generated for a Diebold device. Command 93 is enabled in the AKB NSP's default factory security policy. | |

| 3. | ENABLE REQUIRED SECURITY DEVICE COMMANDS (THALES) | |
|---|---|---|
| a) | The AKDS product requires the following commands to be enabled in the Thales security device(s).<br><br>• Generate RSA Key Set (Command EI)<br>• Load a Secret Key (Command EK)<br>• Generate a MAC on a Public Key (Command EO)<br>• Validate a Certificate and Generate MAC on its Public Key (Command ES)<br>• Generate a Signature (Command EW)<br>• Validate a Signature (Command EY)<br>• Export a DES Key (Command GK)<br>• Hash a Block of Data (Command GM) | |
| End of Section | | |

## NATIVE DIEBOLD 10XX/478X SETUP

### 1. APPLY SCNs

**a)** Apply the SCNs listed in the following files:

| | |
|---|---|
| • Release 6.0 Version 4 | $<VOL>.BA60UD07.SCNDAKDS |
| • Release 6.0 Version 5 | $<VOL>.BA60UD08.SCNLALLD |
| | $<VOL>.BA60UD08.SCNTSSRR |

### 2. MODIFY THE CUSTMACS FILE

**a)** The following Make flag has been added to the CUSTMACS file. Set the flag to TRUE to compile and bind the AKDS extension to the Diebold 10XX/478X device handler.

- atm_dh_c1000_akd_on = $(TRUE)

### 3. MAKE THE SECURITY UTILITIES WITH THE AKDS EXTENSION

**a)** Restore the following AKDS Security Utilities extension subvolume and files.

| Subvolume.File | Type |
|---|---|
| BA60ASRC.BAASRCFM | Make |
| BA60ASRC.BAASRCM | Make |
| BA60ASRC.BAASRCMM | Make |
| BA60ASRC.SECAKDE | TAL External |
| BA60ASRC.SECAKDM | Make |
| BA60ASRC.SECAKDO | TAL object |
| BA60ASRC.SECAKDS | TAL source |

**b)** Run Make on the BA60SRC subvolume. The Make file for this subvolume is BA60SRC.BASRCM. Because the atm_dh_c1000_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS Security Utilities extension will be compiled and bound into the main Security Utilities module.

### 4. MAKE THE DIEBOLD 10XX/478X DH WITH THE AKDS EXTENSION

**a)** Restore the following AKDS extension subvolume and files.

| Subvolume.File | Type |
|---|---|
| AT60A100.ATA100FM | Make |
| AT60A100.ATA100MM | Make |
| AT60A100.C100AKDE | TAL External |
| AT60A100.C100AKDG | TAL source |
| AT60A100.C100AKDM | Make |
| AT60A100.C100AKDO | TAL object |
| AT60A100.C100AKDS | TAL source |

**b)** Run Make on the AT601000 subvolume. The Make file for this subvolume is AT601000.AT1000M. Because the atm_dh_c1000_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS extension will be compiled and bound into the Diebold 10XX/478X device handler.

**c)** Replace the BASE24 system’s current Diebold 10XX/478X device handler object file with the newly compiled object file.

| NATIVE DIEBOLD 10XX/478X SETUP | | |
|---|---|---|
| **5.** | **ADD THE AKDS LCONF PARAMS** | |
| **a)** | Add the following params to the LCONF of each logical network that will support AKDS. | |
| **b)** | **ATM-DH-AKDS-TSS-TIMER**<br><br>The number of seconds the NCR 5XXX device handler process or Diebold 10XX/478X device handler process waits for a reply from the AKDS process before it considers the AKDS request to have timed out. In this case, the requested load is aborted. Valid values are 10–600 seconds. This param defaults to 600 seconds.<br><br>This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in hundredths of a second, to wait for a response from an HSM before the AKDS process considers the request to be timed out. The ATM-DH-AKDS-TSS-TIMER param value must be set to a value large enough to accommodate this length of time plus the time needed to send the request to and receive the response from the AKDS process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window. | |
| **c)** | **ATM-DH-1000-AKDS-SERIAL-NUM-VRFY**<br><br>An optional param indicating whether the Diebold 10XX/478X device handler process verifies the EPP serial number received from the ATM when authenticating and exchanging public keys. If the value matches a valid EPP serial number stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:<br><br>Y = Yes, validate the EPP serial number<br>N = No, do not validate the EPP serial number<br><br>If this param is not configured or contains an invalid value, it defaults to a value of N.<br><br>**Note:** If the value of this param is set to Y, the serial numbers of the Diebold EPPs in the BASE24-atm system must be manually entered on the EPP Serial Number Configuration window before validation can be performed. | |
| **6.** | **CHECK THE BASE24-ATM TERMINAL DATA FILE (ATD) RECORDS** | |
| **a)** | For each Diebold 10XX/478X terminal that supports AKDS, read its record on the BASE24-atm Terminal Data (ATD) file maintenance screen and go to screen 9. | |
| **b)** | The Master Key field should contain a key locator value that links this terminal's ATD record to a corresponding Channel Security Data (CSECD) record in the TSS database. If the Master Key field does not contain a key locator value, change it and update the record.<br><br>**Note:** The key locator value is the same as the ATD record's Term ID field (if the TSS database resides in a single logical network) or a two-byte LN-IND value followed by the Term ID (if the TSS database is partitioned across multiple logical networks). | |
| **c)** | Go to screen 1, and ensure that the NATIVE MESSAGE FORMAT field is set to a 0, to indicate that native Diebold messages are being used. | |
| **d)** | If Release 6.0 Version 10 code or later is being used, go to screen 22, and set the KEY SIGNER field to a reasonable value. The field will default to 'HOST' if nothing is entered. | |

# NATIVE DIEBOLD 10XX/478X SETUP

| | | |
|---|---|---|
| **7.** | **CHECK THE CHANNEL SECURITY DATA (CSECD) RECORDS** | |
| **a)** | For each Diebold 10XX/478X terminal that supports AKDS, read its record in the Channel Security Configuration window. Records are read according to the Channel ID field. The value of the Channel ID must equal the key locator value found in the terminal's BASE24-atm Terminal Data File (ATD) record, screen 9, Master Key field. | |
| **b)** | If a Channel Security Data record does not exist for this terminal, add it. | |
| **c)** | The Key Length of the Exchange Key should be set to **Double**.<br><br>The Key Length of the Inbound Key should be set to **Double**.<br><br>Modify these fields if necessary, save the record, and warmboot all IS processes. | |
| **8.** | **GENERATE AN RSA KEY PAIR FOR THE BASE24 SYSTEM** | |
| **a)** | Contact your Diebold System Engineer to obtain an information packet containing the web address of the certificate authority (Digital Signatures Trust, or DST), your authorization code (password), and a demonstration of the registration website. | |
| **b)** | Generate an RSA public and private key pair for the BASE24 system. This is done one time only for a given BASE24 system unless the RSA key pair is corrupted in the TSS database.<br><br>To do so, go to the Process Control window of the ACI Desktop.<br><br>• Select **XMLS** in the Destination field.<br>• Select **Deliver** in the Command field.<br>• Select **Transaction Security** in the Component field.<br>• Enter **KEYGEN HOST, DIEBOLD, BASE24, BANKNAME, , 2048, , DATE** in the Command Text field.<br><br>**Note: BANKNAME** should be replaced with your own bank name, up to 64 characters.<br><br>**Note: DATE** should be replaced with the effective date required for the Public Keys.<br><br>• **Note: HOST** should be replaced with the value entered in the KEY SIGNER field on screen 22 of the ATD, if Release 6.0 Version 10 code or later is being used. Click on the green arrow in the upper left hand corner of the window to issue the command.<br><br>**Note:** The security device used to generate the RSA key pair is disabled during key generation, which may take up to 90 seconds. Ensure that the security device is PKI-enabled. | |
| **c)** | Access the Certificate Management window on the ACI Desktop to retrieve the BASE24 system public key certificate and format it in the manner that the target certificate authority expects. When the Certificate Management window is displayed, select the Hostname and select Diebold as the device type in the Device Type field. Select the effective date required for the public key. The BASE24 system public key certificate is retrieved and displayed in the Host Public Key Data field. For Diebold, the key is displayed in PKCS#10 Certificate Request format, Base64 encoded. This is the message format required when sending the public key to Diebold's certificate authority. | |

| NATIVE DIEBOLD 10XX/478X SETUP | | |
|---|---|---|
| **d)** | Copy the BASE24 system public key certificate in the Host Public Key Data field and paste it in the appropriate field in the DST-hosted Diebold Host Bank registration pages. Host bank registration involves three phases: authorization, application, and retrieval. For the authorization phase, you must enter a unique authorization code distributed by Diebold to ensure that you are permitted to receive a Host Bank server certificate. Ask your Diebold System Engineer to contact the Diebold Product Manager for their Remote Key product using an e-mail message. The Diebold Product Manager will send you an e-mail message containing the address of the DST-hosted web site, your authorization code (password), and an HTML demonstration of the registration website. An example of this email message is shown below. Each time you need new certificates (e.g., for test, production, and certification) you must contact your Diebold System Engineer to request a new authorization code.<br><br>`This e-mail contains the voucher (Authorization Code) that allows *Financial Institution* to receive host certificates. The information in this message should be regarded with the appropriate level of security, as this voucher retrieves certificates that allow the holder to be identified as a legitimate ATM host by the EPP.`<br><br>`Website: https://secure.digsigtrust.com/tsapp/dbold-ssl-start.jsp`<br><br>`<https://secure.digsigtrust.com/tsapp/dbold-ssl-start.jsp>`<br><br>`Authorization Code: 1234567-XXXX-XXXX`<br><br>`Attached is a demonstration of the host registration site. Please use these test html pages to become familiar with the registration site. Unzip the attached file into a directory before beginning at the file named "Start_Here".`<br><br>For the application phase, you must enter some required and optional contact information (e.g., e-mail address, phone number, and mailing address) for the headquarters of your organization. You must also enter some required and optional contact information about yourself (e.g., your name, job title, and office phone number) and choose a DST passphrase. A DST passphrase is required to renew or revoke your digital certificate, as well as to update your personal information. Finally, you must copy the BASE24 system public key certificate from the Host Public Key Data field on the Certificate Management window and paste it into the CSR field on the DST registration pages. At this point, the application phase is complete. DST verifies the BASE24 system public key certificate and creates two certificates of the BASE24 system public key, signed by the certificate authority's primary and secondary private keys, respectively. The certificate authority also creates two additional certificates. One certificate contains the primary certificate authority's public signature verification key, signed by the certificate authority's primary private key. The second certificate contains the secondary certificate authority's public signature verification key, signed by the certificate authority's secondary private key.<br><br>For the retrieval phase, you must copy the following certificates generated or displayed by DST in your browser and paste them in the appropriate fields on the Certificate Management window in the following steps. All certificates are in PKCS #7 Cryptographic Message format. The following table shows you which field on the Certificate Management window to paste each certificate from DST and in which of the following steps you paste the certificate. | |

## NATIVE DIEBOLD 10XX/478X SETUP

| DST Certificate | Certificate Management Field |
|---|---|
| DST Root CA X3 Certificate | Generate Message Digest > Root Certificate Authority Certificate (step e)<br>and/or<br>Validate Certificates/Signatures > Root Certificate Authority Certificate (step h) |
| DST Primary Sub-CA Certificate | Validate Certificates/Signatures > CA Primary Certificate (step h) |
| DST Secondary Sub-CA Certificate | Validate Certificates/Signatures > CA Secondary Certificate (step h) |
| Primary Host Bank Server Certificate | Validate Certificates/Signatures > Host Primary Certificate (step h) |
| Secondary Host Bank Server Certificate | Validate Certificates/Signatures > Host Secondary Certificate (step h) |

| | | |
|---|---|---|
| **e)** | If you are using a Thales e-Security HSM, you may skip to step h.<br>If you are using an Atalla NSP, create a message digest of the root certificate authority certificate (DST Root CA X3 Certificate). Copy the root certificate authority certificate returned from DST into the Root Certificate Authority Certificate field on the Generate Message Digest tab of the Certificate Management window and click Save. You can click Expand View to expand this field if needed. The NSP uses command 126 to create a message digest value that is displayed in the Message Digest field on this tab. | |
| **f)** | For an Atalla NSP only, encrypt the value returned in the Message Digest field from the previous step under variant 30 of the MFK using the SCT or SCA. Variant 30 is automatically assumed by the SCT or SCA The SCT requires the message digest to be input as two 16-character fields. Use the Utilities Menu > Encrypt Challenge function on the SCT to encrypt the message digest value. You must input the message digest as two 16 byte values (Left Challenge and Right Challenge). Using the SCA instead, with the SCA connected, enter a card and PIN. Select security management. Select Calculate AKB/Cryptogram. For Cryptogram Type select Challenge/Response and select Next. Enter the leftmost 16 characters and select Next. When the next screen is displayed select OK and put in a second Security Administrator card and enter its PIN. Select Confirm. The response and check digits will be displayed. Select Save to save the response in the Data Table. Repeat the process for the 16 rightmost characters. The results are two 16 byte values that comprise the root certificate hash value. | |
| **g)** | Copy or enter the 32 byte (16 byte left and 16 byte right) values from the previous step into the Root Certificate Hash Value field on the Validate Certificates/Signatures tab. | |
| **h)** | Copy the root certificate authority certificate, primary and secondary certificate authority's public signature verification key certificates, and the primary and secondary BASE24 system public signature verification key certificates returned from DST in step d into the | |

<table>
<tr><th colspan="3">NATIVE DIEBOLD 10XX/478X SETUP</th></tr>
<tr><td></td><td>appropriate fields on the Validate Certificates/Signatures tab of the Certificate Management window and click Save. When you click Save, the following processing is performed.<br><br>1) The root certificate authority certificate is validated using the hash value input from step g (for Atalla NSPs only), the self-signed signature of the certificate is validated, and the root certificate authority public key is stored in the Host Public Key Security File (HSPKD).<br><br>2) The signatures of the primary and secondary certificate authority’s public signature verification keys are validated using the root certificate authority public key stored in step h-1. The primary certificate authority public key is stored in the HSPKD. The secondary certificate authority public key is stored in the Host Public Key Security File (HSPKD).<br><br>3) The signatures of the BASE24 system (host) primary and secondary certificates are validated using the primary and secondary certificate authority’s public signature verification keys stored in step h-2, respectively. The primary BASE24 system public key signature is stored in the HSPKD. The secondary BASE24 system public key signature is stored in the HSPKD.<br><br>Note: The certificates for the certificate authority are parsed and only the public key values are stored in the HSPKD. For the BASE24 system (host) public keys, the full certificate signatures are stored in the HSPKD.</td><td></td></tr>
<tr><td>i)</td><td>Once the above steps are completed, the following key data is entered into the database for the AKDS process or in the tamper-resistant memory of the hardware security module.<br><br>• Root certificate authority public key<br>• Primary and secondary BASE24 system public key signatures (signed by the certificate authority’s primary and secondary private keys, respectively)<br>• Primary and secondary certificate authority’s public signature verification keys<br>• The BASE24 system’s private signature generation key</td><td></td></tr>
<tr><td>9.</td><td colspan="2">DISABLE THE KEYGEN COMMAND (OPTIONAL)</td></tr>
<tr><td></td><td>After you have successfully generated an RSA key pair for the BASE24 system and have had the public key signed by the certificate authority, you may optionally disable use of the KEYGEN command. This helps prevent the unintentional generation of a new RSA key pair, which, if it occurred, would require you to repeat the public key signing process. The KEYGEN command can be re-enabled if it is needed at a later time.</td><td></td></tr>
<tr><td>a)</td><td>Go to the Disable KEYGEN window of the ACI Desktop.</td><td></td></tr>
<tr><td>b)</td><td>Select HOST in the Host Name field.<br><br>Note: HOST can be replaced with a unique name if Release 6.0 Version 10 code or later is being used. Ensure that the name being used matches the one used in the KEYGEN command.<br><br>Select Diebold in the Device Type field.<br><br>Select KEYGEN command is disabled in the KEYGEN Command Status field.</td><td></td></tr>
<tr><td>c)</td><td>Save the record.</td><td></td></tr>
<tr><th colspan="3">End of Section</th></tr>
</table>

| NATIVE DIEBOLD 10XX/478X TERMINAL DOWNLOADS | | |
|---|---|---|
| **1.** | **AUTHENTICATE AND EXCHANGE THE PUBLIC KEYS** | |
| | Before you can automatically download an ATM master key to the EPP of an ATM, the EPP and the BASE24 system must authenticate each other and exchange their public keys. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD PUBLIC KEY field. | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the authentication and public key exchange. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^C1000, “9500LOADPUBLIC <terminal ID>” | |
| **2.** | **DOWNLOAD THE MASTER KEY** | |
| | Once the EPP has been successfully authenticated and has exchanged public keys with the BASE24 system, you can securely download the master key to the terminal. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD MASTER KEY field | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the master key download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^C1000, “9500LOADMASTER <terminal ID>” | |
| **3.** | **DOWNLOAD THE MAC KEY AND/OR COMMUNICATIONS KEY** | |
| | Once the master key has been successfully downloaded to the terminal, you can securely download the MAC key and/or communications key, encrypted under the new master key. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | If you are downloading a new MAC key:<br><br>• Position the cursor in the LOAD MAC KEY field.<br><br>• Press the F1 key. | |

| NATIVE DIEBOLD 10XX/478X TERMINAL DOWNLOADS | | |
|---|---|---|
| **d)** | If you are downloading a new communications key:<br>• Enter a value of ‘2’ in the LOAD PIN KEY field.<br>• Press the F1 key while the cursor is positioned in the LOAD PIN KEY field. | |
| **e)** | Check the event log for the successful completion of the download. | |
| **4.** | **DOWNLOAD ALL KEYS (OPTIONAL)** | |
| | Optionally, you may load all of the aforementioned keys (public, master, MAC and communication keys) using a single DCT command, the LOAD ALL KEYS command. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD ALL KEYS field. | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br>>DELIVER PROCESS P1A^C1000, “9500LOADALLKEYS <terminal ID>” | |
| **End of Section** | | |

| NATIVE NCR 5XXX BASIC SCHEME SETUP | | |
|---|---|---|
| **1.** | **APPLY SCNs** | |
| **a)** | Apply the SCNs listed in the following files:<br><br>• Release 6.0 Version 3 $<VOL>.BA60UD06.SCNNAKDS<br>• Release 6.0 Version 5 $<VOL>.BA60UD08.SCNLALLN<br>$<VOL>.BA60UD08.SCNTSSRR | |
| **2.** | **MODIFY THE CUSTMACS FILE** | |
| **a)** | The following Make flag has been added to the CUSTMACS file. Set the flag to TRUE to compile and bind the AKDS extension to the NCR 5XXX device handler.<br><br>• atm_dh_n50_akd_on = $(TRUE) | |
| **3.** | **MAKE THE SECURITY UTILITIES WITH THE AKDS EXTENSION** | |
| **a)** | Restore the following AKDS Security Utilities extension subvolume and files.<br><br><u>Subvolume.File</u> <u>Type</u><br>BA60ASRC.BAASRCFM Make<br>BA60ASRC.BAASRCM Make<br>BA60ASRC.BAASRCMM Make<br>BA60ASRC.SECAKDE TAL External<br>BA60ASRC.SECAKDM Make<br>BA60ASRC.SECAKDO TAL object<br>BA60ASRC.SECAKDS TAL source | |
| **b)** | Run Make on the BA60SRC subvolume. The Make file for this subvolume is BA60SRC.BASRCM. Because the atm_dh_n50_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS Security Utilities extension will be compiled and bound into the main Security Utilities module. | |
| **4.** | **MAKE THE NCR 5XXX DH WITH THE AKDS EXTENSION** | |
| **a)** | Restore the following AKDS extension subvolume and files.<br><br><u>Subvolume.File</u> <u>Type</u><br>AT60AN50.ATAN50FM Make<br>AT60AN50.ATAN50MM Make<br>AT60AN50.N50AKDE TAL External<br>AT60AN50.N50AKDG TAL source<br>AT60AN50.N50AKDM Make<br>AT60AN50.N50AKDO TAL object<br>AT60AN50.N50AKDS TAL source | |
| **b)** | Run Make on the AT60N50 subvolume. The Make file for this subvolume is AT60N50.ATN50M. Because the atm_dh_n50_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS extension will be compiled and bound into the NCR 5XXX device handler. | |
| **c)** | Replace the BASE24 system's current NCR device handler object file with the newly compiled object file. | |

| NATIVE NCR 5XXX BASIC SCHEME SETUP | | |
|---|---|---|
| **5.** | **ADD THE AKDS LCONF PARAMS** | |
| **a)** | Add the following params to the LCONF of each logical network that will support AKDS. | |
| **b)** | **ATM-DH-AKDS-TSS-TIMER**<br><br>The number of seconds the NCR 5XXX device handler process or Diebold 10XX/478X device handler process waits for a reply from the AKDS process before it considers the AKDS request to have timed out. In this case, the requested load is aborted. Valid values are 10–600 seconds. This param defaults to 600 seconds.<br><br>This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in hundredths of a second, to wait for a response from an HSM before the AKDS process considers the request to be timed out. The ATM-DH-AKDS-TSS-TIMER param value must be set to a value large enough to accommodate this length of time plus the time needed to send the request to and receive the response from the AKDS process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window. | |
| **c)** | **ATM-DH-N50-AKDS-SERIAL-NUM-VRFY**<br><br>An optional param indicating whether the NCR 5XXX device handler process verifies the EPP serial number received from the ATM when authenticating and exchanging public keys. If the value matches a valid EPP serial number stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:<br><br>Y = Yes, validate the EPP serial number<br>N = No, do not validate the EPP serial number<br><br>If this param is not configured or contains an invalid value, it defaults to a value of N.<br><br>**Note:** If the value of this param is set to Y, the serial numbers of the NCR EPPs in the BASE24-atm system must be manually entered on the EPP Serial Number Configuration window before validation can be performed. | |
| **6.** | **CHECK THE BASE24-ATM TERMINAL DATA FILE (ATD) RECORDS** | |
| **a)** | For each NCR 5XXX terminal that supports AKDS, read its record on the BASE24-atm Terminal Data (ATD) file maintenance screen and go to screen 9. | |
| **b)** | The Master Key field should contain a key locator value that links this terminal's ATD record to a corresponding Channel Security Data (CSECD) record in the TSS database. If the Master Key field does not contain a key locator value, change it and update the record.<br><br>**Note:** The key locator value is the same as the ATD record's Term ID field (if the TSS database resides in a single logical network) or a two-byte LN-IND value followed by the Term ID (if the TSS database is partitioned across multiple logical networks). | |
| **c)** | Go to screen 1, and ensure that the NATIVE MESSAGE FORMAT field is set to a 0, to indicate that native NCR messages are being used. | |
| **d)** | If Release 6.0 Version 10 code or later is being used, go to screen 22, and set the KEY SIGNER field to a reasonable value. The field will default to 'HOST' if nothing is entered. | |

<table>
<tr><th colspan="3">NATIVE NCR 5XXX BASIC SCHEME SETUP</th></tr>
<tr><th>7.</th><th colspan="2">CHECK THE CHANNEL SECURITY DATA (CSECD) RECORDS</th></tr>
<tr><td>a)</td><td>For each NCR 5XXX terminal that supports AKDS, read its record in the Channel Security Configuration window. Records are read according to the Channel ID field. The value of the Channel ID must equal the key locator value found in the terminal's BASE24-atm Terminal Data File (ATD) record, screen 9, Master Key field.</td><td></td></tr>
<tr><td>b)</td><td>If a Channel Security Data record does not exist for this terminal, add it.</td><td></td></tr>
<tr><td>c)</td><td>The Key Length of the Exchange Key should be set to Double.<br><br>The Key Length of the Inbound Key should be set to Double.<br><br>The MAC option must be set to Pseudo Triple DES Macing with 32 bits. TSS 6.2 and above supports this for Thales and Eracom, Atalla has had this support all along.<br><br>Modify these fields if necessary, save the record, and warmboot all IS processes.</td><td></td></tr>
<tr><th>8.</th><th colspan="2">GENERATE AN RSA KEY PAIR FOR THE BASE24 SYSTEM</th></tr>
<tr><td>a)</td><td>To start the process of submitting a public key and getting it signed by NCR, contact your NCR/Reseller account manager. The account manager will then establish your name, mailing address, e-mail address, telephone and fax numbers and advise NCR Customer and Sales Support to send you the following forms:<br><br>• Signature Request Process (NCR_001)<br>• NCR Approved Contacts (NCR_002)<br><br>This form will be completed by NCR before it is sent to you.<br><br>• Signature Request (NCR_003)</td><td></td></tr>
<tr><td>b)</td><td>Generate an RSA public and private key pair for the BASE24 system. This is done one time only for a given BASE24 system unless the RSA key pair is corrupted in the TSS database.<br><br>To do so, go to the Process Control window of the ACI Desktop.<br><br>• Select XMLS in the Destination field.<br>• Select Deliver in the Command field.<br>• Select Transaction Security in the Component field.<br>• Enter KEYGEN HOST, NCR, , , , 2048, , DATE! in the Command Text field.<br>Note: HOST should be replaced with the value entered in the KEY SIGNER field on screen 22 of the ATD, if Release 6.0 Version 10 code or later is being used.<br><br>Note: DATE should be replaced with the effective data required for the Public Key<br><br>• Click on the green arrow in the upper left hand corner of the window to issue the command.<br><br>Note: The security device used to generate the RSA key pair is disabled during key generation, which may take up to 90 seconds. Ensure that the security device is PKI-enabled.</td><td></td></tr>
</table>

## NATIVE NCR 5XXX BASIC SCHEME SETUP

**c)** Access the Certificate Management window on the ACI Desktop to retrieve the BASE24 system public key and format it in the manner that the NCR trusted center expects. When the Certificate Management window is displayed, select the Host name, and select the NCR device type in the Device Type field. Select the effective date required for the public keys. The BASE24 system public key is retrieved, signed, formatted, and displayed in the Host Public Key Data field. A 40-byte SHA-1 hash value is also generated and displayed in the SHA-1 Hash Value field. The BASE24 system public key is self-signed and displayed in ASN.1 format. This is the message format required when sending the public key to the NCR trusted center.

**d)** Copy the BASE24 system public key cryptogram in the Host Public Key Data field, format it as required, and send it on the appropriate media (e.g, on a diskette or CD-ROM in a tamper-resistant envelope) to your primary NCR contact specified on form NCR_002. Follow the format and procedures prescribed on form NCR_001.

At the same time, complete form NCR_003 and send it by e-mail message or by facsimile (fax) to all Key Signing Individuals (i.e., the primary and all secondary NCR contacts specified on form NCR_002). Make sure you enter the 40-byte hash value displayed in the SHA-1 Hash Value field on the Host Public Key Data tab in the SHA-1 (Fields 1–6) field on form NCR_003.

**e)** The NCR trusted center validates the BASE24 system public key signature and SHA-1 hash value. The Key Signing Individual will send an e-mail message to you to acknowledge receipt of the diskette or CD-ROM and to indicate whether the signature and hash value are correct. If either are incorrect, you must resend both form NCR_003 and the BASE24 system public key. If both are correct, the trusted center will sign the BASE24 system public key, generate SHA-1 hash values for both the BASE24 system public key signature and the self-signed root NCR public key, and return them back to you on a diskette or CD-ROM delivered by a courier service. NCR will also complete and issue the following two forms by e-mail message or by facsimile (fax) to the approved customer contact. You must use these forms to verify that the SHA-1 hash values over the self-signed root NCR public key and BASE24 system public key signature are valid.

- NCR Public Key (NCR_004)
- NCR Signature of Customer Public Key (NCR_005)

The diskette or CD-ROM returned to you will contain several files. Each file contains a key or hash value. You must copy the contents of some of these files and paste them into fields on the Certificate Management window in subsequent steps as shown in the following table.

| Value (NCR Filename) | Certificate Management Field |
|---|---|
| Self-signed root NCR public key (PK_NCR.txt) | Generate Message Digest > Certificate Authority Public Key (step f-1)<br><br>and<br><br>Validate Certificates/Signatures > Certificate Authority Public Key (step i-1) |

## NATIVE NCR 5XXX BASIC SCHEME SETUP

| | |
|---|---|
| SHA-1 hash of self-signed NCR public key (PK_NCR_digest.txt) | Generate Message Digest > SHA-1 Hash Value (step f-2) |
| BASE24 system public key signature (PK_(cust)_ncr_sig) | Validate Certificates/Signatures > Host Signature (step i-1) |
| SHA-1 hash of BASE24 system public key (PK_(cust)_NCR_sig_digest.txt) | Validate Certificates/Signatures > SHA-1 Hash Value (step i-2) |

**f)** On the Generate Message Digest tab of the Certificate Management window, perform the following:

1) Copy the self-signed root NCR public key cryptogram returned on diskette or CD-ROM from the NCR trusted center into the Certificate Authority Public Key field. You can click Expand View to expand this field if needed.
2) Enter the SHA-1 hash value of the self-signed NCR public key into the SHA-1 Hash Value field.
3) Click Save.

If you are using an Atalla NSP, the NSP validates the signature of the self-signed root NCR public key, validates the SHA-1 hash value for the key, and generates a message digest of the self-signed root NCR public key, which is displayed in the Message Digest field.

If you are using a Thales e-Security HSM, the HSM validates the signature and SHA-1 hash value of the self-signed root NCR public key. Thales HSMs do not create a message digest.

Any validation errors are reported in screen messages on the Certificate Management window. If the key signature or SHA-1 hash value are incorrect, you must request NCR to resend the self-signed root NCR public key, SHA-1 hash value, and form NCR_004. The NSP uses command 126 to create a message digest value that is displayed in the Message Digest field on this tab.

**g)** If you are using a Thales e-Security HSM, you may skip this step.

For an Atalla NSP only, encrypt the value returned in the Message Digest field from the previous step under variant 30 of the MFK using the SCT or SCA. Variant 30 is automatically assumed by both the SCT and SCA. Both the SCT and SCA require the message digest to be input as two 16-character fields. On the SCT, use the Utilities Menu > Encrypt Challenge function to encrypt the message digest value. On the SCA, use the Calculate AKB/Cryptogram menu (Challenge/Response) to encrypt the message digest value. You must input the message digest as two 16 byte values (Left Challenge and Right Challenge on the SCT or Component block 1 and Component block 2 on the SCA). The results are two 16 byte values that comprise the root certificate hash value.

**h)** If you are using a Thales e-Security HSM, you may skip this step.

For an Atalla NSP only, copy or enter the 32 byte (16 byte left and 16 byte right) values from the previous step into the Root Certificate Hash Value field on the Validate Certificates/Signatures tab.

<table>
<tr><th colspan="3">NATIVE NCR 5XXX BASIC SCHEME SETUP</th></tr>
<tr><td>i)</td><td>On the Validate Certificates/Signatures tab of the Certificate Management window, perform the following:<br>1) Copy the self-signed root NCR public key and BASE24 system public key signature returned from the NCR trusted center into the Certificate Authority Public Key and Host Signature fields, respectively, on the Validate Certificates/Signatures tab of the Certificate Management window. You can click Expand View to expand either of these fields if needed.<br>2) Enter the SHA-1 hash value for the BASE24 system public key signature into the SHA-1 Hash Value field.<br>3) Click Save.<br><br>The Atalla NSP or Thales HSM validates the signature and SHA-1 hash value of the BASE24 system public key received from NCR. If validated successfully, the BASE24 system public key signature is stored in the Host Public Key Security File (HSPKD).<br>For an Atalla NSP only, the self-signed root NCR public key is validated using the hash value input from step h, imported, and stored in the HSPKD along with the message digest.<br>Any validation errors are reported in screen messages on the Certificate Management window. If the key signature or SHA-1 hash value are incorrect, you must request NCR to resend the BASE24 system public key signature, SHA-1 hash value, and form NCR_005.</td><td></td></tr>
<tr><td>j)</td><td>Once the above steps are completed, the following key data is now entered into the database for the AKDS process.<br>• The BASE24 system’s public and private keys<br>• The signature of the BASE24 system’s public key signed by the root NCR private key<br>• The self-signed root NCR public key<br>• A MAC of the self-signed root NCR public key (for Thales e-Security HSMs only)</td><td></td></tr>
<tr><td>9.</td><td colspan="2">DISABLE THE KEYGEN COMMAND (OPTIONAL)</td></tr>
<tr><td></td><td>After you have successfully generated an RSA key pair for the BASE24 system and have had the public key signed by the certificate authority, you may optionally disable use of the KEYGEN command. This helps prevent the unintentional generation of a new RSA key pair, which, if it occurred, would require you to repeat the public key signing process. The KEYGEN command can be re-enabled if it is needed at a later time.</td><td></td></tr>
<tr><td>a)</td><td>Go to the Disable KEYGEN window of the ACI Desktop.</td><td></td></tr>
<tr><td>b)</td><td>Select HOST in the Host Name field.<br>Note: HOST can be replaced with a unique name if Release 6.0 Version 10 code or later is being used. Ensure that the name being used matches the one used in the KEYGEN command.<br>Select NCR in the Device Type field.<br><br>Select KEYGEN command is disabled in the KEYGEN Command Status field.</td><td></td></tr>
<tr><td>c)</td><td>Save the record.</td><td></td></tr>
<tr><td colspan="3">End of Section</td></tr>
</table>

# NATIVE NCR 5XXX BASIC TERMINAL DOWNLOADS

| | | |
|---|---|---|
| **1.** | **DOWNLOAD THE KEY ENTRY MODE** | |
| | This procedure can be performed if you wish to switch an EPP operating in Basic Alpha PIN pad and Encryptor (BAPE) emulation mode to full EPP mode (PKI-enabled).<br>**Warning:** This command causes all DES keys in the encryptor to be deleted.<br><br>This procedure is not necessary if the key entry mode was set through supervisor mode at the terminal itself. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Enter one of the following numbers in the LOAD KEY ENTRY MODE field.<br><br>1 Single length without XOR<br>2 Single length with XOR<br>3 Double length with XOR<br>4 Double length restricted<br><br>**Note:** NCR recommends that double-length keys be used when implementing AKDS. Furthermore, double-length keys are required for Triple DES. That said, **Load Key Entry Mode option 3 (double length with XOR) is recommended**. | |
| **d)** | Press the F6 key while the cursor is positioned in the LOAD KEY ENTRY MODE field. | |
| **e)** | Check the event log for the successful completion of the key entry mode download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, “9500LKEYMODE <terminal ID> 3” | |
| **2.** | **AUTHENTICATE AND EXCHANGE THE PUBLIC KEYS** | |
| | Before you can automatically download an ATM master key to the EPP of an ATM, the EPP and the BASE24 system must authenticate each other and exchange their public keys. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD PUBLIC KEY field. | |
| **d)** | Press the F6 key. | |
| **e)** | Check the event log for the successful completion of the authentication and public key exchange. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, “9500LPUBLIC <terminal ID>” | |

| NATIVE NCR 5XXX BASIC TERMINAL DOWNLOADS | | |
|---|---|---|
| **3.** | **DOWNLOAD THE MASTER KEY** | |
| | Once the EPP has been successfully authenticated and has exchanged public keys with the BASE24 system, and the key entry mode has been set, you can securely download the master key to the terminal. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD MASTER KEY field | |
| **d)** | Press the F6 key. | |
| **e)** | Check the event log for the successful completion of the master key download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, “9500LMASTER <terminal ID>” | |
| **4.** | **DOWNLOAD THE MAC KEY AND/OR COMMUNICATIONS KEY** | |
| | Once the master key has been successfully downloaded to the terminal, you can securely download the MAC key and/or communications key, encrypted under the new master key. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | If you are downloading a new MAC key:<br><br>• Position the cursor in the LOAD MAC KEY field.<br><br>• Press the F6 key. | |
| **d)** | If you are downloading a new communications key:<br><br>• Enter a value of ‘2’ in the LOAD KEY TYPE field.<br><br>• Press the F6 key while the cursor is positioned in the LOAD KEY TYPE field. | |
| **e)** | Check the event log for the successful completion of the download. | |
| **5.** | **DOWNLOAD ALL KEYS (OPTIONAL)** | |
| | Optionally, you may load all of the aforementioned keys (public, master, MAC and communication keys) using a single DCT command, the LOAD ALL KEYS command. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |

| NATIVE NCR 5XXX BASIC TERMINAL DOWNLOADS | | |
|---|---|---|
| **c)** | Position the cursor in the LOAD ALL KEYS field. | |
| **d)** | Press the F6 key. | |
| **e)** | Check the event log for the successful completion of the download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, “9500LALLKEYS <terminal ID>” | |
| **End of Section** | | |

# NATIVE NCR 5XXX ENHANCED SCHEME SETUP

## 1. APPLY SCNs

**a)** Apply the SCNs listed in the following files:

| | |
|---|---|
| • Release 6.0 Version 3 | $<VOL>.BA60UD06.SCNNAKDS |
| • Release 6.0 Version 5 | $<VOL>.BA60UD08.SCNLALLN<br>$<VOL>.BA60UD08.SCNTSSRR |
| • Release 6.0 Version 10 | $<VOL>.BA60UD0D.SCNTSS |
| • Release 6.0 Version post 10 | $<VOL>.BA60UD0F.SCNRKL |

## 2. MODIFY THE CUSTMACS FILE

**a)** The following Make flag has been added to the CUSTMACS file. Set the flag to TRUE to compile and bind the AKDS extension to the NCR 5XXX device handler.

- atm_dh_n50_akd_on = $(TRUE)

## 3. MAKE THE SECURITY UTILITIES WITH THE AKDS EXTENSION

**a)** Restore the following AKDS Security Utilities extension subvolume and files.

| Subvolume.File | Type |
|---|---|
| BA60ASRC.BAASRCFM | Make |
| BA60ASRC.BAASRCM | Make |
| BA60ASRC.BAASRCMM | Make |
| BA60ASRC.SECAKDE | TAL External |
| BA60ASRC.SECAKDM | Make |
| BA60ASRC.SECAKDO | TAL object |
| BA60ASRC.SECAKDS | TAL source |

**b)** Run Make on the BA60SRC subvolume. The Make file for this subvolume is BA60SRC.BASRCM. Because the atm_dh_n50_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS Security Utilities extension will be compiled and bound into the main Security Utilities module.

## 4. MAKE THE NCR 5XXX DH WITH THE AKDS EXTENSION

**a)** Restore the following AKDS extension subvolume and files.

| Subvolume.File | Type |
|---|---|
| AT60AN50.ATAN50FM | Make |
| AT60AN50.ATAN50MM | Make |
| AT60AN50.N50AKDE | TAL External |
| AT60AN50.N50AKDG | TAL source |
| AT60AN50.N50AKDM | Make |
| AT60AN50.N50AKDO | TAL object |
| AT60AN50.N50AKDS | TAL source |

| NATIVE NCR 5XXX ENHANCED SCHEME SETUP | | |
|---|---|---|
| **b)** | Run Make on the AT60N50 subvolume. The Make file for this subvolume is AT60N50.ATN50M. Because the atm_dh_n50_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS extension will be compiled and bound into the NCR 5XXX device handler. | |
| **c)** | Replace the BASE24 system's current NCR device handler object file with the newly compiled object file. | |
| **5.** | **ADD THE AKDS LCONF PARAMS** | |
| **a)** | Add the following params to the LCONF of each logical network that will support AKDS. | |
| **b)** | **ATM-DH-AKDS-TSS-TIMER**<br><br>The number of seconds the NCR 5XXX device handler process or Diebold 10XX/478X device handler process waits for a reply from the AKDS process before it considers the AKDS request to have timed out. In this case, the requested load is aborted. Valid values are 10–600 seconds. This param defaults to 600 seconds.<br><br>This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in hundredths of a second, to wait for a response from an HSM before the AKDS process considers the request to be timed out. The ATM-DH-AKDS-TSS-TIMER param value must be set to a value large enough to accommodate this length of time plus the time needed to send the request to and receive the response from the AKDS process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window. | |
| **c)** | **ATM-DH-N50-AKDS-SERIAL-NUM-VRFY**<br><br>An optional param indicating whether the NCR 5XXX device handler process verifies the EPP serial number received from the ATM when authenticating and exchanging public keys. If the value matches a valid EPP serial number stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:<br><br>Y = Yes, validate the EPP serial number<br>N = No, do not validate the EPP serial number<br><br>If this param is not configured or contains an invalid value, it defaults to a value of N.<br><br>**Note:** If the value of this param is set to Y, the serial numbers of the NCR EPPs in the BASE24-atm system must be manually entered on the EPP Serial Number Configuration window before validation can be performed. | |

| | | |
|---|---|---|
| **6.** | **CHECK THE BASE24-ATM TERMINAL DATA FILE (ATD) RECORDS** | |
| **a)** | For each NCR 5XXX terminal that supports AKDS, read its record on the BASE24-atm Terminal Data (ATD) file maintenance screen and go to screen 9. | |
| **b)** | The Master Key field should contain a key locator value that links this terminal's ATD record to a corresponding Channel Security Data (CSECD) record in the TSS database. If the Master Key field does not contain a key locator value, change it and update the record.<br>**Note:** The key locator value is the same as the ATD record's Term ID field (if the TSS database resides in a single logical network) or a two-byte LN-IND value followed by the Term ID (if the TSS database is partitioned across multiple logical networks). | |
| **c)** | Go to screen 1, and ensure that the NATIVE MESSAGE FORMAT field is set to a 0, to indicate that native NCR messages are being used. | |
| **d)** | If Release 6.0 Version 10 code or later is being used, go to screen 22, and set the KEY SIGNER field to a reasonable value. The field will default to 'HOST' if nothing is entered. | |
| **7.** | **CHECK THE CHANNEL SECURITY DATA (CSECD) RECORDS** | |
| **a)** | For each NCR 5XXX terminal that supports AKDS, read its record in the Channel Security Configuration window. Records are read according to the Channel ID field. The value of the Channel ID must equal the key locator value found in the terminal's BASE24-atm Terminal Data File (ATD) record, screen 9, Master Key field. | |
| **b)** | If a Channel Security Data record does not exist for this terminal, add it. | |
| **c)** | The Key Length of the Exchange Key should be set to **Double**.<br>The Key Length of the Inbound Key should be set to **Double**.<br>The MAC option must be set to **Pseudo Triple DES Macing with 32 bits**. TSS 6.2 and above supports this for Thales and Eracom, Atalla has had this support all along.<br>Modify these fields if necessary, save the record, and warmboot all IS processes. | |
| **8.** | **GENERATE AN RSA KEY PAIR FOR THE BASE24 SYSTEM** | |
| **a)** | To start the process of submitting a public key and getting it signed by NCR, contact your NCR/Reseller account manager. The account manager will then establish your name, mailing address, e-mail address, telephone and fax numbers and advise NCR Customer and Sales Support to send you the following forms:<br>• Signature Request Process (NCR_001)<br>• NCR Approved Contacts (NCR_002) | |

| | | |
|---|---|---|
| | This form will be completed by NCR before it is sent to you.<br><br>• Signature Request (NCR_003) | |
| **b)** | Generate RSA public and private key pairs for the BASE24 system. This is done one time only for a given BASE24 system unless the RSA key pair is corrupted in the TSS database.<br><br>To do so, go to the Process Control window of the ACI Desktop.<br><br>• Select **XMLS** in the Destination field.<br>• Select **Deliver** in the Command field.<br>• Select **Transaction Security** in the Component field.<br>• Enter **KEYGEN HOST, NCR, , , , 2048, B, DATE!** in the Command Text field.<br>**Note: HOST** should be replaced with the value entered in the KEY SIGNER field on screen 22 of the ATD, if Release 6.0 Version 10 code or later is being used. .<br><br>**Note: DATE** should be replaced with the effective data required for the Public Key<br><br>• Click on the green arrow in the upper left hand corner of the window to issue the command.<br><br>**Note:** The security device used to generate the RSA key pair is disabled during key generation, which may take up to 90 seconds. Ensure that the security device is PKI-enabled. | |
| **c)** | Access the Certificate Management window on the ACI Desktop to retrieve the root BASE24 system public key and format it in the manner that the NCR trusted center expects. When the Certificate Management window is displayed, select the Host name, and enter NCR device type in the Device Type field. Select the effective date used to generate the key. The root BASE24 system public key is retrieved, signed, formatted, and displayed in the Host Public Key Data field. A 40-byte SHA-1 hash value is also generated and displayed in the SHA-1 Hash Value field. The root BASE24 system public key is self-signed and displayed in ASN.1 format. This is the message format required when sending the public key to the NCR trusted center. | |
| **d)** | Copy the root BASE24 system public key cryptogram in the Root Host Public Key Data field, format it as required, and send it on the appropriate media (e.g, on a diskette or CD-ROM in a tamper-resistant envelope) to your primary NCR contact specified on form NCR_002. Follow the format and procedures prescribed on form NCR_001.<br><br>At the same time, complete form NCR_003 and send it by e-mail message or by facsimile (fax) to all Key Signing Individuals (i.e., the primary and all secondary NCR contacts specified on form NCR_002). Make sure you enter the 40-byte hash value displayed in the SHA-1 Hash Value field on the Root Host Public Key Data tab in the SHA-1 (Fields 1–6) field on form NCR_003. | |

**e)**

The NCR trusted center validates the root BASE24 system public key signature and SHA-1 hash value. The Key Signing Individual will send an e-mail message to you to acknowledge receipt of the diskette or CD-ROM and to indicate whether the signature and hash value are correct. If either are incorrect, you must resend both form NCR_003 and the root BASE24 system public key. If both are correct, the trusted center will sign the root BASE24 system public key, generate SHA-1 hash values for both the root BASE24 system public key signature and the self-signed root NCR public key, and return them back to you on a diskette or CD-ROM delivered by a courier service. NCR will also complete and issue the following two forms by e-mail message or by facsimile (fax) to the approved customer contact. You must use these forms to verify that the SHA-1 hash values over the self-signed root NCR public key and root BASE24 system public key signature are valid.

- NCR Public Key (NCR_004)
- NCR Signature of Customer Public Key (NCR_005)

The diskette or CD-ROM returned to you will contain several files. Each file contains a key or hash value. You must copy the contents of some of these files and paste them into fields on the Certificate Management window in subsequent steps as shown in the following table.

| **Value (NCR Filename)** | **Certificate Management Field** |
|---|---|
| Self-signed root NCR public key (PK_NCR.txt) | Generate Message Digest > Certificate Authority Public Key (step f-1)<br><br>and<br><br>Validate Certificates/Signatures > Certificate Authority Public Key (step i-1) |
| SHA-1 hash of self-signed NCR public key (PK_NCR_digest.txt) | Generate Message Digest > SHA-1 Hash Value (step f-2) |
| Root BASE24 system public key signature (PK_(cust)_ncr_sig) | Validate Certificates/Signatures > Host Signature (step i-1) |
| SHA-1 hash of root BASE24 system public key (PK_(cust)_NCR_sig_digest.txt) | Validate Certificates/Signatures > SHA-1 Hash Value (step i-2) |

**f)**

On the Generate Message Digest tab of the Certificate Management window, perform the following:

1) Copy the self-signed root NCR public key cryptogram returned on diskette or CD-ROM from the NCR trusted center into the Certificate Authority Public Key field. You can click Expand View to expand this field if needed.

2) Enter the SHA-1 hash value of the self-signed NCR public key into the SHA-1 Hash Value field.

3) Click Save.

If you are using an Atalla NSP, the NSP validates the signature of the self-signed root

| | | |
|---|---|---|
| | NCR public key, validates the SHA-1 hash value for the key, and generates a message digest of the self-signed root NCR public key, which is displayed in the Message Digest field.<br><br>If you are using a Thales e-Security HSM, the HSM validates the signature and SHA-1 hash value of the self-signed root NCR public key. Thales HSMs do not create a message digest.<br><br>Any validation errors are reported in screen messages on the Certificate Management window. If the key signature or SHA-1 hash value are incorrect, you must request NCR to resend the self-signed root NCR public key, SHA-1 hash value, and form NCR_004. The NSP uses command 126 to create a message digest value that is displayed in the Message Digest field on this tab. | |
| **g)** | If you are using a Thales e-Security HSM, you may skip this step.<br><br>For an Atalla NSP only, encrypt the value returned in the Message Digest field from the previous step under variant 30 of the MFK using the SCT or SCA. Variant 30 is automatically assumed by both the SCT and SCA. Both the SCT and SCA require the message digest to be input as two 16-character fields. On the SCT, use the Utilities Menu > Encrypt Challenge function to encrypt the message digest value. On the SCA, use the Calculate AKB/Cryptogram menu (Challenge/Response) to encrypt the message digest value. You must input the message digest as two 16 byte values (Left Challenge and Right Challenge on the SCT or Component block 1 and Component block 2 on the SCA). The results are two 16 byte values that comprise the root certificate hash value. | |
| **h)** | If you are using a Thales e-Security HSM, you may skip this step.<br><br>For an Atalla NSP only, copy or enter the 32 byte (16 byte left and 16 byte right) values from the previous step into the Root Certificate Hash Value field on the Validate Certificates/Signatures tab. | |
| **i)** | On the Validate Certificates/Signatures tab of the Certificate Management window, perform the following:<br><br>1) Copy the self-signed root NCR public key and root BASE24 system public key signature returned from the NCR trusted center into the Certificate Authority Public Key and Host Signature fields, respectively, on the Validate Certificates/Signatures tab of the Certificate Management window. You can click Expand View to expand either of these fields if needed.<br><br>2) Enter the SHA-1 hash value for the root BASE24 system public key signature into the SHA-1 Hash Value field.<br><br>3) Click Save.<br><br>The Atalla NSP or Thales HSM validates the signature and SHA-1 hash value of the root BASE24 system public key received from NCR. If validated successfully, the root BASE24 system public key signature is stored in the Host Public Key Security File (HSPKD).<br><br>For an Atalla NSP only, the self-signed root NCR public key is validated using the hash value input from step h, imported, and stored in the HSPKD along with the message digest. | |

| | | |
|---|---|---|
| | Any validation errors are reported in screen messages on the Certificate Management window. If the key signature or SHA-1 hash value are incorrect, you must request NCR to resend the BASE24 system public key signature, SHA-1 hash value, and form NCR_005. | |
| **j)** | Once the above steps are completed, the following key data is now entered into the database for the AKDS process.<br>• The root BASE24 system's public and private keys<br>• The signature of the root BASE24 system's public key signed by the root NCR private key<br>• The self-signed root NCR public key<br>• A MAC of the self-signed root NCR public key (for Thales e-Security HSMs only) | |
| **9.** | **DISABLE THE KEYGEN COMMAND (OPTIONAL)** | |
| | After you have successfully generated an RSA key pair for the BASE24 system and have had the public key signed by the certificate authority, you may optionally disable use of the KEYGEN command. This helps prevent the unintentional generation of a new RSA key pair, which, if it occurred, would require you to repeat the public key signing process. The KEYGEN command can be re-enabled if it is needed at a later time. | |
| **a)** | Go to the Disable KEYGEN window of the ACI Desktop. | |
| **b)** | Select **HOST** in the Host Name field.<br>**Note: HOST can be replaced with a unique name if Release 6.0 Version 10 code or later is being used. Ensure that the name being used matches the one used in the KEYGEN command.**<br>Select **NCR** in the Device Type field.<br><br>Select **KEYGEN command is disabled** in the KEYGEN Command Status field. | |
| **c)** | Save the record. | |
| | **End of Section** | |

| NATIVE NCR 5XXX ENHANCED TERMINAL DOWNLOADS | | |
|---|---|---|
| **1.** | **DOWNLOAD THE KEY ENTRY MODE** | |
| | This procedure can be performed if you wish to switch an EPP operating in Basic Alpha PIN pad and Encryptor (BAPE) emulation mode to full EPP mode (PKI-enabled).<br>If you wish to change the key mode for other reasons, it must be performed after the authentication and exchange public keys step (step 2 below).<br>**Warning:** This command causes all DES keys in the encryptor to be deleted.<br><br>This procedure is not necessary if the key entry mode was set through supervisor mode at the terminal itself. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Enter one of the following numbers in the LOAD KEY ENTRY MODE field.<br><br>1 Single length without XOR<br>2 Single length with XOR<br>3 Double length with XOR<br>4 Double length restricted<br><br>**Note:** NCR recommends that double-length keys be used when implementing AKDS. Furthermore, double-length keys are required for Triple DES. That said, **Load Key Entry Mode option 3 (double length with XOR) is recommended**. | |
| **d)** | Press the F6 key while the cursor is positioned in the LOAD KEY ENTRY MODE field. | |
| **e)** | Check the event log for the successful completion of the key entry mode download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, “9500LKEYMODE <terminal ID> 3” | |
| **2.** | **AUTHENTICATE AND EXCHANGE THE PUBLIC KEYS** | |
| | Before you can automatically download an ATM master key to the EPP of an ATM, the EPP and the BASE24 system must authenticate each other and exchange their public keys. If this step has already been performed previously for this ATM, (for the Enhanced Scheme) a delete of the root BASE24 system’s public key must be performed before the authentication and exchange of the public keys can be repeated. See step 5 below for the delete command. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |

| NATIVE NCR 5XXX ENHANCED TERMINAL DOWNLOADS | | |
|---|---|---|
| **c)** | Position the cursor in the LOAD PUBLIC KEY field. | |
| **d)** | Press the F6 key. | |
| **e)** | Check the event log for the successful completion of the authentication and public key exchange. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, “9500LPUBLIC <terminal ID>” | |
| **3.** | **DOWNLOAD THE MASTER KEY** | |
| | Once the EPP has been successfully authenticated and has exchanged public keys with the BASE24 system, and the key entry mode has been set, you can securely download the master key to the terminal. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD MASTER KEY field | |
| **d)** | Press the F6 key. | |
| **e)** | Check the event log for the successful completion of the master key download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, “9500LMASTER <terminal ID>” | |
| **4.** | **DOWNLOAD THE MAC KEY AND/OR COMMUNICATIONS KEY** | |
| | Once the master key has been successfully downloaded to the terminal, you can securely download the MAC key and/or communications key, encrypted under the new master key. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | If you are downloading a new MAC key:<br>• Position the cursor in the LOAD MAC KEY field.<br>• Press the F6 key. | |
| **d)** | If you are downloading a new communications key:<br>• Enter a value of ‘2’ in the LOAD KEY TYPE field.<br>• Press the F6 key while the cursor is positioned in the LOAD KEY TYPE field. | |
| **e)** | Check the event log for the successful completion of the download. | |

| | | |
|---|---|---|
| **5.** | **DELETE PUBLIC KEY(S)** | |
| | Before an authentication and exchange of public keys step can be repeated, a delete of the root BASE24 system's public key must be performed. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Enter one of the following values in the DELETE PUBLIC KEY value:<br><br>'R' – Delete the root BASE24 system public key (also deletes the primary BASE24 system public key)<br>'B' – Delete the primary BASE24 system public key only<br><br>**Note**: 'R' is recommended | |
| **d)** | Press the F6 key while the cursor is positioned in front of the DELETE PUBLIC KEY field. | |
| **e)** | Check the event log for the successful completion of the download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, "9500DELPUBLIC <x> <terminal ID>"<br><br>where <x> is 'R' or 'P'. | |
| **6.** | **DOWNLOAD ALL KEYS (OPTIONAL)** | |
| | Optionally, you may load all of the aforementioned keys (public, master, MAC and communication keys) using a single DCT command, the LOAD ALL KEYS command. This command will now perform a delete of the root BASE24 system public key before it loads any keys.<br><br>**Note**: If this is the first time that AKDS is being performed for an ATM configured as Enhanced scheme, the LOADALLKEYS will fail, as no public keys exist in the EPP to be deleted. In this unique case, perform the LOADPUBLIC, LOADMASTER and LOAD of the communication keys as separate steps for this first time only. Subsequently LOADALLKEYS can be used. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD ALL KEYS field. | |
| **d)** | Press the F6 key. | |
| **e)** | Check the event log for the successful completion of the download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, "9500LALLKEYS <terminal ID>" | |
| **End of Section** | | |

| NCR 5XXX CERTIFICATES SETUP | | |
|---|---|---|
| **1.** | **APPLY SCNs** | |
| **a)** | Apply the SCNs listed in the following file:<br><br>• Release 6.0 Version 6 $<VOL>.BA60UD09.SCNNCRT | |
| **2.** | **MODIFY THE CUSTMACS FILE** | |
| **a)** | The following Make flag already exists in the CUSTMACS file. Ensure that the flag is set to TRUE to compile and bind the AKDS extension to the NCR 5XXX device handler.<br><br>• atm_dh_n50_akd_on = $(TRUE) | |
| **3.** | **MAKE THE SECURITY UTILITIES WITH THE AKDS EXTENSION** | |
| **a)** | Restore the following AKDS Security Utilities extension subvolume and files.<br><br><u>Subvolume.File</u> — <u>Type</u><br>BA60ASRC.BAASRCFM — Make<br>BA60ASRC.BAASRCM — Make<br>BA60ASRC.BAASRCMM — Make<br>BA60ASRC.SECAKDE — TAL External<br>BA60ASRC.SECAKDM — Make<br>BA60ASRC.SECAKDO — TAL object<br>BA60ASRC.SECAKDS — TAL source | |
| **b)** | Run Make on the BA60SRC subvolume. The Make file for this subvolume is BA60SRC.BASRCM. Because the atm_dh_n50_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS Security Utilities extension will be compiled and bound into the main Security Utilities module. | |
| **4.** | **MAKE THE NCR 5XXX DH WITH THE AKDS EXTENSION** | |
| **a)** | Restore the following AKDS extension subvolume and files.<br><br><u>Subvolume.File</u> — <u>Type</u><br>AT60AN50.ATAN50FM — Make<br>AT60AN50.ATAN50MM — Make<br>AT60AN50.N50AKDE — TAL External<br>AT60AN50.N50AKDG — TAL source<br>AT60AN50.N50AKDM — Make<br>AT60AN50.N50AKDO — TAL object<br>AT60AN50.N50AKDS — TAL source | |
| **b)** | Run Make on the AT60N50 subvolume. The Make file for this subvolume is AT60N50.ATN50M. Because the atm_dh_n50_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS extension will be compiled and bound into the NCR 5XXX device handler. | |
| **c)** | Replace the BASE24 system’s current NCR device handler object file with the newly compiled object file. | |

# NCR 5XXX CERTIFICATES SETUP

| | | |
|---|---|---|
| **5.** | **ADD THE AKDS LCONF PARAMS** | |
| **a)** | Ensure that the following params exist in the LCONF of each logical network that will support AKDS. | |
| **b)** | **ATM-DH-AKDS-TSS-TIMER**<br>The number of seconds the NCR 5XXX device handler process or Diebold 10XX/478X device handler process waits for a reply from the AKDS process before it considers the AKDS request to have timed out. In this case, the requested load is aborted. Valid values are 10–600 seconds. This param defaults to 600 seconds.<br>This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in hundredths of a second, to wait for a response from an HSM before the AKDS process considers the request to be timed out. The ATM-DH-AKDS-TSS-TIMER param value must be set to a value large enough to accommodate this length of time plus the time needed to send the request to and receive the response from the AKDS process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window. | |
| **c)** | **ATM-DH-N50-AKDS-SERIAL-NUM-VRFY**<br>An optional param indicating whether the NCR 5XXX device handler process verifies the EPP serial number received from the ATM when authenticating and exchanging public keys. If the value matches a valid EPP serial number stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:<br><br>Y = Yes, validate the EPP serial number<br>N = No, do not validate the EPP serial number<br>If this param is not configured or contains an invalid value, it defaults to a value of N.<br>**Note:** If the value of this param is set to Y, the serial numbers of the NCR EPPs in the BASE24-atm system must be manually entered on the EPP Serial Number Configuration window before validation can be performed. | |
| **6.** | **CHECK THE BASE24-ATM TERMINAL DATA FILE (ATD) RECORDS** | |
| **a)** | For each Diebold terminal that supports AKDS, read its record on the BASE24-atm Terminal Data (ATD) file maintenance screen and go to screen 9. | |
| **b)** | The Master Key field should contain a key locator value that links this terminal's ATD record to a corresponding Channel Security Data (CSECD) record in the TSS database. If the Master Key field does not contain a key locator value, change it and update the record.<br><br>**Note:** The key locator value is the same as the ATD record's Term ID field (if the TSS database resides in a single logical network) or a two-byte LN-IND value followed by the Term ID (if the TSS database is partitioned across multiple logical networks). | |
| **c)** | Go to screen 1, and ensure that the NATIVE MESSAGE FORMAT field is set to a 0, to indicate that native NCR messages are being used. | |
| **d)** | If Release 6.0 Version 10 code or later is being used, go to screen 22, and set the KEY SIGNER field to a reasonable value. The field will default to 'HOST' if nothing is entered. | |

| NCR 5XXX CERTIFICATES SETUP | | |
|---|---|---|
| **7.** | **CHECK THE CHANNEL SECURITY DATA (CSECD) RECORDS** | |
| **a)** | For each Diebold terminal that supports AKDS, read its record in the Channel Security Configuration window. Records are read according to the Channel ID field. The value of the Channel ID must equal the key locator value found in the terminal's BASE24-atm Terminal Data File (ATD) record, screen 9, Master Key field. | |
| **b)** | If a Channel Security Data record does not exist for this terminal, add it. | |
| **c)** | The Key Length of the Exchange Key should be set to **Double**.<br><br>The Key Length of the Inbound Key should be set to **Double**.<br><br>Modify these fields if necessary, save the record, and warmboot all IS processes. | |
| **8.** | **GENERATE AN RSA KEY PAIR FOR THE BASE24 SYSTEM** | |
| **a)** | As you are using Diebold ATMs for this solution, follow the directions in section 8 of the Native Diebold 10XX/478X Setup Section. | |
| **9.** | **DISABLE THE KEYGEN COMMAND (OPTIONAL)** | |
| | After you have successfully generated an RSA key pair for the BASE24 system and have had the public key signed by the certificate authority, you may optionally disable use of the KEYGEN command. This helps prevent the unintentional generation of a new RSA key pair, which, if it occurred, would require you to repeat the public key signing process. The KEYGEN command can be re-enabled if it is needed at a later time. | |
| **a)** | Go to the Disable KEYGEN window of the ACI Desktop. | |
| **b)** | Select **HOST** in the Host Name field.<br><br>**Note: HOST can be replaced with a unique name if Release 6.0 Version 10 code or later is being used. Ensure that the name being used matches the one used in the KEYGEN command.**<br><br>Select **Diebold** in the Device Type field.<br><br>Select **KEYGEN command is disabled** in the KEYGEN Command Status field. | |
| **c)** | Save the record. | |
| **End of Section** | | |

| NCR 5XXX CERTIFICATES TERMINAL DOWNLOADS | | |
|---|---|---|
| **1.** | **AUTHENTICATE AND EXCHANGE THE PUBLIC KEYS** | |
| | Before you can automatically download an ATM master key to the EPP of an ATM, the EPP and the BASE24 system must authenticate each other and exchange their public keys. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. (The NCR screen is used because the NCR 5XXX device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD PUBLIC KEY field. | |
| **d)** | Press the F6 key. | |
| **e)** | Check the event log for the successful completion of the authentication and public key exchange. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, "9500LPUBLIC <terminal ID>" | |
| **2.** | **DOWNLOAD THE MASTER KEY** | |
| | Once the EPP has been successfully authenticated and has exchanged public keys with the BASE24 system, and the key entry mode has been set, you can securely download the master key to the terminal. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. (The NCR screen is used because the NCR 5XXX device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD MASTER KEY field | |
| **d)** | Press the F6 key. | |
| **e)** | Check the event log for the successful completion of the master key download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, "9500LMASTER <terminal ID>" | |
| **3.** | **DOWNLOAD THE MAC KEY AND/OR COMMUNICATIONS KEY** | |
| | Once the master key has been successfully downloaded to the terminal, you can securely download the MAC key and/or communications key, encrypted under the new master key. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |

| NCR 5XXX CERTIFICATES TERMINAL DOWNLOADS | | |
|---|---|---|
| **c)** | If you are downloading a new MAC key:<br>• Position the cursor in the LOAD MAC KEY field.<br>• Press the F6 key. | |
| **d)** | If you are downloading a new communications key:<br>• Enter a value of '2' in the LOAD KEY TYPE field.<br>• Press the F6 key while the cursor is positioned in the LOAD KEY TYPE field. | |
| **e)** | Check the event log for the successful completion of the download. | |
| **4.** | **DOWNLOAD ALL KEYS (OPTIONAL)** | |
| | Optionally, you may load all of the aforementioned keys (public, master, MAC and communication keys) using a single DCT command, the LOAD ALL KEYS command. | |
| **a)** | Go to Device Control Terminal (DCT) screen 45 and position the cursor in the DOWNLINE LOAD DATA field. Press the F1 key. The NCR 5XXX LOAD COMMANDS screen will appear. (The NCR screen is used because the NCR 5XXX device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD ALL KEYS field. | |
| **d)** | Press the F6 key. | |
| **e)** | Check the event log for the successful completion of the download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br>>DELIVER PROCESS P1A^N50, "9500LALLKEYS <terminal ID>" | |
| **End of Section** | | |

| DIEBOLD 912 SIGNATURES SETUP | | |
|---|---|---|
| **1.** | **APPLY SCNs** | |
| **a)** | Apply the SCNs listed in the following file:<br><br>• Release 6.0 Version 8 $<VOL>.BA60UD0B.SCNDAKD | |
| **2.** | **MODIFY THE CUSTMACS FILE** | |
| **a)** | The following Make flag already exists in the CUSTMACS file. Ensure that the flag is set to TRUE to compile and bind the AKDS extension to the Diebold 10XX/478X device handler.<br><br>• atm_dh_c1000_akd_on = $(TRUE) | |
| **3.** | **MAKE THE SECURITY UTILITIES WITH THE AKDS EXTENSION** | |
| **c)** | Restore the following AKDS Security Utilities extension subvolume and files.<br><br><u>Subvolume.File</u> — <u>Type</u><br>BA60ASRC.BAASRCFM — Make<br>BA60ASRC.BAASRCM — Make<br>BA60ASRC.BAASRCMM — Make<br>BA60ASRC.SECAKDE — TAL External<br>BA60ASRC.SECAKDM — Make<br>BA60ASRC.SECAKDO — TAL object<br>BA60ASRC.SECAKDS — TAL source | |
| **d)** | Run Make on the BA60SRC subvolume. The Make file for this subvolume is BA60SRC.BASRCM. Because the atm_dh_c1000_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS Security Utilities extension will be compiled and bound into the main Security Utilities module. | |
| **4.** | **MAKE THE DIEBOLD 10XX/478X DH WITH THE AKDS EXTENSION** | |
| **d)** | Restore the following AKDS extension subvolume and files.<br><br><u>Subvolume.File</u> — <u>Type</u><br>AT60A100.ATA100FM — Make<br>AT60A100.ATA100MM — Make<br>AT60A100.C100AKDE — TAL External<br>AT60A100.C100AKDG — TAL source<br>AT60A100.C100AKDM — Make<br>AT60A100.C100AKDO — TAL object<br>AT60A100.C100AKDS — TAL source | |
| **e)** | Run Make on the AT601000 subvolume. The Make file for this subvolume is AT601000.AT1000M. Because the atm_dh_c1000_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS extension will be compiled and bound into the Diebold 10XX/478X device handler. | |
| **f)** | Replace the BASE24 system's current Diebold 10XX/478X device handler object file with the newly compiled object file. | |

| DIEBOLD 912 SIGNATURES SETUP | | |
|---|---|---|
| **5.** | **ADD THE AKDS LCONF PARAMS** | |
| **a)** | Ensure that the following params exist in the LCONF of each logical network that will support AKDS. | |
| **b)** | **ATM-DH-AKDS-TSS-TIMER**<br><br>The number of seconds the NCR 5XXX device handler process or Diebold 10XX/478X device handler process waits for a reply from the AKDS process before it considers the AKDS request to have timed out. In this case, the requested load is aborted. Valid values are 10–600 seconds. This param defaults to 600 seconds.<br><br>This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in hundredths of a second, to wait for a response from an HSM before the AKDS process considers the request to be timed out. The ATM-DH-AKDS-TSS-TIMER param value must be set to a value large enough to accommodate this length of time plus the time needed to send the request to and receive the response from the AKDS process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window. | |
| **c)** | **ATM-DH-1000-AKDS-SERIAL-NUM-VRFY**<br><br>An optional param indicating whether the Diebold 10XX/478X device handler process verifies the EPP serial number received from the ATM when authenticating and exchanging public keys. If the value matches a valid EPP serial number stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:<br><br>Y = Yes, validate the EPP serial number<br>N = No, do not validate the EPP serial number<br><br>If this param is not configured or contains an invalid value, it defaults to a value of N.<br><br>**Note:** If the value of this param is set to Y, the serial numbers of the NCR EPPs in the BASE24-atm system must be manually entered on the EPP Serial Number Configuration window before validation can be performed. | |
| **6.** | **CHECK THE BASE24-ATM TERMINAL DATA FILE (ATD) RECORDS** | |
| **a)** | For each NCR terminal that supports AKDS, read its record on the BASE24-atm Terminal Data (ATD) file maintenance screen and go to screen 9. | |
| **b)** | The Master Key field should contain a key locator value that links this terminal's ATD record to a corresponding Channel Security Data (CSECD) record in the TSS database. If the Master Key field does not contain a key locator value, change it and update the record.<br><br>**Note:** The key locator value is the same as the ATD record's Term ID field (if the TSS database resides in a single logical network) or a two-byte LN-IND value followed by the Term ID (if the TSS database is partitioned across multiple logical networks). | |
| **c)** | Go to screen 1, and ensure that the NATIVE MESSAGE FORMAT field is set to a 0, to indicate that native Diebold messages are being used. | |
| **d)** | If Release 6.0 Version 10 code or later is being used, go to screen 22, and set the KEY SIGNER field to a reasonable value. The field will default to 'HOST' if nothing is entered. | |

| DIEBOLD 912 SIGNATURES SETUP | | |
|---|---|---|
| **7.** | **CHECK THE CHANNEL SECURITY DATA (CSECD) RECORDS** | |
| **a)** | For each NCR terminal that supports AKDS, read its record in the Channel Security Configuration window. Records are read according to the Channel ID field. The value of the Channel ID must equal the key locator value found in the terminal's BASE24-atm Terminal Data File (ATD) record, screen 9, Master Key field. | |
| **b)** | If a Channel Security Data record does not exist for this terminal, add it. | |
| **c)** | The Key Length of the Exchange Key should be set to **Double**.<br><br>The Key Length of the Inbound Key should be set to **Double**.<br><br>Modify these fields if necessary, save the record, and warmboot all IS processes. | |
| **8.** | **GENERATE AN RSA KEY PAIR FOR THE BASE24 SYSTEM** | |
| **b)** | As you are using NCR ATMs for this solution, follow the directions in section 8 of the Native NCR 5XXX Setup Section. | |
| **9.** | **DISABLE THE KEYGEN COMMAND (OPTIONAL)** | |
| | After you have successfully generated an RSA key pair for the BASE24 system and have had the public key signed by the certificate authority, you may optionally disable use of the KEYGEN command. This helps prevent the unintentional generation of a new RSA key pair, which, if it occurred, would require you to repeat the public key signing process. The KEYGEN command can be re-enabled if it is needed at a later time. | |
| **a)** | Go to the Disable KEYGEN window of the ACI Desktop. | |
| **b)** | Select **HOST** in the Host Name field.<br><br>**Note: HOST can be replaced with a unique name if Release 6.0 Version 10 code or later is being used. Ensure that the name being used matches the one used in the KEYGEN command.**<br><br>Select **Diebold** in the Device Type field.<br><br>Select **KEYGEN command is disabled** in the KEYGEN Command Status field. | |
| **c)** | Save the record. | |
| **End of Section** | | |

| DIEBOLD 912 SIGNATURES TERMINAL DOWNLOADS | | |
|---|---|---|
| **1.** | **AUTHENTICATE AND EXCHANGE THE PUBLIC KEYS** | |
| | Before you can automatically download an ATM master key to the EPP of an ATM, the EPP and the BASE24 system must authenticate each other and exchange their public keys. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. (The Diebold screen is used because the Diebold 10XX/478X device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD PUBLIC KEY field. | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the authentication and public key exchange. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^C1000, “9500LOADPUBLIC <terminal ID>” | |
| **2.** | **DOWNLOAD THE MASTER KEY** | |
| | Once the EPP has been successfully authenticated and has exchanged public keys with the BASE24 system, you can securely download the master key to the terminal. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. (The Diebold screen is used because the Diebold 10XX/478X device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD MASTER KEY field | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the master key download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^C1000, “9500LOADMASTER <terminal ID>” | |
| **3.** | **DOWNLOAD THE MAC KEY AND/OR COMMUNICATIONS KEY** | |
| | Once the master key has been successfully downloaded to the terminal, you can securely download the MAC key and/or communications key, encrypted under the new master key. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | If you are downloading a new MAC key:<br><br>• Position the cursor in the LOAD MAC KEY field. | |

| DIEBOLD 912 SIGNATURES TERMINAL DOWNLOADS | | |
|---|---|---|
| | • Press the F1 key. | |
| **d)** | If you are downloading a new communications key:<br><br>• Enter a value of ‘2’ in the LOAD KEY TYPE field.<br><br>• Press the F1 key while the cursor is positioned in the LOAD KEY TYPE field. | |
| **e)** | Check the event log for the successful completion of the download. | |
| **4.** | **DOWNLOAD ALL KEYS (OPTIONAL)** | |
| | Optionally, you may load all of the aforementioned keys (public, master, MAC and communication keys) using a single DCT command, the LOAD ALL KEYS command. | |
| **f)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. (The Diebold screen is used because the Diebold 10XX/478X device handler is being used). | |
| **g)** | Enter the ATM TERMINAL ID. | |
| **h)** | Position the cursor in the LOAD ALL KEYS field. | |
| **i)** | Press the F1 key. | |
| **j)** | Check the event log for the successful completion of the download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^C1000, “9500LOADALLKEYS <terminal ID>” | |
| **End of Section** | | |

| **SHARED 912 SIGNATURES SETUP** | | |
|---|---|---|
| **1.** | **APPLY SCNs** | |
| **a)** | Apply the SCNs listed in the following file:<br>• Release 6.0 Version 6 $<VOL>.BA60UD09.SCNWDAK<br>If NCR ATMs are to be installed running Wincor's 912 Signature software, apply the SCNs listed in the following file:<br>• Release 6.0 Version 8 $<VOL>.BA60UD0B.SCNWAKD | |
| **2.** | **MODIFY THE CUSTMACS FILE** | |
| **a)** | The following Make flag has been added to the CUSTMACS file. Set the flag to TRUE to compile and bind the Shared AKDS extension to the Diebold 10XX/478X device handler.<br>• atm_dh_c1000_shrd_akd_on = $(TRUE) | |
| **3.** | **MAKE THE SECURITY UTILITIES WITH THE AKDS EXTENSION** | |
| **a)** | Restore the following AKDS Security Utilities extension subvolume and files.<br><br><u>Subvolume.File</u> <u>Type</u><br>BA60ASRC.BAASRCFM Make<br>BA60ASRC.BAASRCM Make<br>BA60ASRC.BAASRCMM Make<br>BA60ASRC.SECAKDE TAL External<br>BA60ASRC.SECAKDM Make<br>BA60ASRC.SECAKDO TAL object<br>BA60ASRC.SECAKDS TAL source | |
| **b)** | Run Make on the BA60SRC subvolume. The Make file for this subvolume is BA60SRC.BASRCM. Because the atm_dh_c1000_shrd_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS Security Utilities extension will be compiled and bound into the main Security Utilities module. | |
| **4.** | **MAKE THE DIEBOLD 10XX/478X DH WITH THE SHARED AKDS EXTENSION** | |
| **a)** | Restore the following AKDS extension subvolume and files.<br><br><u>Subvolume.File</u> <u>Type</u><br>AT60AW10.ATAW10FM Make<br>AT60AW10.ATAW10MM Make<br>AT60AW10.C10WAKDE TAL External<br>AT60AW10.C10WAKDG TAL source<br>AT60AW10.C10WAKDM Make<br>AT60AW10.C10WAKDO TAL object<br>AT60AW10.C10WAKDS TAL source | |
| **b)** | Run Make on the AT601000 subvolume. The Make file for this subvolume is AT601000.AT1000M. Because the atm_dh_c1000_shrd_akd_on flag was set to TRUE in the CUSTMACS file, the Shared AKDS extension will be compiled and bound into the Diebold 10XX/478X device handler. | |

| SHARED 912 SIGNATURES SETUP | | |
|---|---|---|
| **c)** | Replace the BASE24 system's current Diebold 10XX/478X device handler object file with the newly compiled object file. | |
| **5.** | **ADD THE AKDS LCONF PARAMS** | |
| **a)** | Add the following params to the LCONF of each logical network that will support AKDS. | |
| **b)** | **ATM-DH-AKDS-TSS-TIMER**<br><br>The number of seconds the NCR 5XXX device handler process or Diebold 10XX/478X device handler process waits for a reply from the AKDS process before it considers the AKDS request to have timed out. In this case, the requested load is aborted. Valid values are 10–600 seconds. This param defaults to 600 seconds.<br><br>This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in hundredths of a second, to wait for a response from an HSM before the AKDS process considers the request to be timed out. The ATM-DH-AKDS-TSS-TIMER param value must be set to a value large enough to accommodate this length of time plus the time needed to send the request to and receive the response from the AKDS process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window. | |
| **c)** | **ATM-DH-1000-AKDS-SERIAL-NUM-VRFY**<br><br>An optional param indicating whether the Diebold 10XX/478X device handler process verifies the EPP serial number received from the ATM when authenticating and exchanging public keys. If the value matches a valid EPP serial number stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:<br><br>Y = Yes, validate the EPP serial number<br>N = No, do not validate the EPP serial number<br><br>If this param is not configured or contains an invalid value, it defaults to a value of N.<br><br>**Note:** If the value of this param is set to Y, the serial numbers of the NCR or Wincor EPPs in the BASE24-atm system must be manually entered on the EPP Serial Number Configuration window before validation can be performed. | |
| **6.** | **CHECK THE BASE24-ATM TERMINAL DATA FILE (ATD) RECORDS** | |
| **a)** | For each NCR or Wincor terminal that supports AKDS, read its record on the BASE24-atm Terminal Data (ATD) file maintenance screen and go to screen 9. | |
| **b)** | The Master Key field should contain a key locator value that links this terminal's ATD record to a corresponding Channel Security Data (CSECD) record in the TSS database. If the Master Key field does not contain a key locator value, change it and update the record.<br><br>**Note:** The key locator value is the same as the ATD record's Term ID field (if the TSS database resides in a single logical network) or a two-byte LN-IND value followed by the Term ID (if the TSS database is partitioned across multiple logical networks). | |
| **c)** | Go to screen 1, and ensure that the NATIVE MESSAGE FORMAT field is set to a 1, to indicate that Wincor specification messages are being used. | |

| **SHARED 912 SIGNATURES SETUP** | | |
|---|---|---|
| **d)** | Go to screen 22, and set the KEY SIGNER field to a reasonable value. The field will default to 'HOST' if nothing is entered. | |
| **7.** | **CHECK THE CHANNEL SECURITY DATA (CSECD) RECORDS** | |
| **a)** | For each NCR or Wincor terminal that supports AKDS, read its record in the Channel Security Configuration window. Records are read according to the Channel ID field. The value of the Channel ID must equal the key locator value found in the terminal's BASE24-atm Terminal Data File (ATD) record, screen 9, Master Key field. | |
| **b)** | If a Channel Security Data record does not exist for this terminal, add it. | |
| **c)** | The Key Length of the Exchange Key should be set to **Double**.<br><br>The Key Length of the Inbound Key should be set to **Double**.<br><br>For NCR ATMs, the MAC option must be set to **Pseudo Triple DES Macing with 32 bits**. TSS 6.2 and above supports this for Thales and Eracom, Atalla has had this support all along.<br><br>Modify these fields if necessary, save the record, and warmboot all IS processes. | |
| **8.** | **GENERATE AN RSA KEY PAIR FOR THE BASE24 SYSTEM** | |
| **a)** | If you are using **NCR ATMs**, follow the directions in section 8 of the Native NCR 5XXX Setup Section. **Note:** In the **Keygen** command**, HOST** should be replaced with the value entered in the KEY SIGNER field on screen 22 of the ATD, and the device type to be used is WINCORNCR instead of NCR.<br>If the customer already has an NCR public key established in their environment, they will need to copy that key and change the implementation type to Wincor NCR:<br><br>In DALCI:<br>Select * from host_public_key where pk_impl == 0,insertform;<br><br>Change the pk_impl to '3' in both records and copy them back into the file via dalci obey<br><br>If you are using **Wincor ATMs**, follow the directions here.<br><br>To start the process of submitting a public key and getting it signed by Wincor Nixdorf, contact your Wincor Nixdorf representative and request a Key Form.<br><br>**Note:** The procedures are essentially the same whether you are generating an RSA key pair for a test or production environment on the BASE24 system. For testing, a test EPP provided by Wincor Nixdorf must be used instead of a production EPP. When you initially contact Wincor Nixdorf to start the process of submitting a public key and getting it signed by Wincor Nixdorf, you must specify whether the key to be signed is for testing or production. You should use the same forms and procedures for getting both test and production keys. Wincor Nixdorf Customer Key Center – International (CKCI) signs test keys using a root public key that is different from the one it uses to sign production keys. | |
| **b)** | Generate an RSA public and private key pair for the BASE24 system. This is done one time only for a given BASE24 system unless the RSA key pair is corrupted in the TSS database.<br><br>To do so, go to the Process Control window of the ACI Desktop. | |

<table>
<tr><th colspan="3">SHARED 912 SIGNATURES SETUP</th></tr>
<tr><td></td><td>
<ul>
<li>Select XMLS in the Destination field.</li>
<li>Select Deliver in the Command field.</li>
<li>Select Transaction Security in the Component field.</li>
<li>Enter KEYGEN HOST, WINCORSIG, , , , 2048, , DATE ! in the Command Text field.<br><br>Note: HOST should be replaced with the value entered in the KEY SIGNER field on screen 22 of the ATD.<br><br>Note: DATE should be replaced with the effective data required for the Public Key.</li>
<li>Click on the green arrow in the upper left hand corner of the window to issue the command.</li>
</ul>
Note: The security device used to generate the RSA key pair is disabled during key generation, which may take up to 90 seconds. Ensure that the security device is PKI-enabled.
</td><td></td></tr>
<tr><td>c)</td><td>Access the Certificate Management window on the ACI Desktop to retrieve the BASE24 system public key certificate and format it in the manner expected by the Wincor Nixdorf Customer Key Center – International (CKCI). When the Certificate Management window is displayed, select the Hostname, and Wincor Signature as the device type in the Device Type field. Select the effective date required for the public key. The BASE24 system public key certificate is retrieved and displayed in the Host Public Key Data field. A 40-byte SHA-1 hash value is also generated and displayed in the SHA-1 Hash Value field. The BASE24 system public key is displayed in ASN.1 format. This is the message format required when sending the public key to CKCI.</td><td></td></tr>
<tr><td>d)</td><td>Enter the 40-byte hash value displayed in the SHA-1 Hash Value field on the Host Public Key Data tab in the Digest (SHA-1) of Customer’s Public Key field on the Wincor Nixdorf Key Form. Choose a Key Length of 2048 bits, and an Exchange Format of ASCII Hex. Choosing the ASCII Hex option ensures that the signature of your BASE24 system public key returned by Wincor Nixdorf will be in the correct format to enter into the ACI Desktop. After completing the remainder of the Key Form, fax it to the Wincor Nixdorf Customer Key Center – International (CKCI).<br><br>When requested by Wincor Nixdorf, copy the BASE24 system public key cryptogram from the Host Public Key Data field on the Certificate Management window into a text file. Send the text file on the appropriate media to Wincor Nixdorf following the format and procedures requested by the Wincor Nixdorf Customer Key Center – International (CKCI).<br><br>The Wincor Nixdorf Customer Key Center – International (CKCI) validates the SHA-1 hash value for your BASE24 system. Wincor Nixdorf will contact you to indicate whether the hash value is correct. If it is incorrect, you must resend the BASE24 system public key and hash value. If it is correct, the Wincor Nixdorf Customer Key Center – International (CKCI) creates a signature of the BASE24 system public key, generates a SHA-1 hash value for the root Wincor Nixdorf public key, and returns them back to you in the media format you requested. You will receive two files. Each file contains a key or hash value in ASCII Hex format. You must copy the contents of these files and paste them into fields on the Certificate Management window in subsequent steps as shown in the following table.</td><td></td></tr>
</table>

# SHARED 912 SIGNATURES SETUP

| Value (Wincor Nixdorf Filename) | Certificate Management Field |
|---|---|
| Root Wincor Nixdorf public key pkSignWN<br><br>(pkSignWN<cust>T.mpk if this is a test key, or pkSignWN<cust>P.mpk if this is a production key) | Generate Message Digest > Certificate Authority Public Key (step e.1)<br><br>and<br><br>Validate Certificates/Signatures > Certificate Authority Public Key (step i.1) |
| SHA-1 hash of Wincor Nixdorf public key<br><br>Digest public key, calculated with SHA1<br><br>(pkSignWN<cust>T.mpk if this is for a test key, or pkSignWN<cust>P.mpk if this is for a production key) | Generate Message Digest > SHA-1 Hash Value (step e.2) |
| BASE24 system public key Signature<br><br>(siHSM<cust>T.sig if this is for a test key, or siHSM<cust>P.sig if this is for a production key) | Validate Certificates/Signatures >Host Signature (step i.2) |

**e)** On the Generate Message Digest tab of the Certificate Management window, perform the following:

1) Copy the root Wincor Nixdorf public key cryptogram from the Wincor Nixdorf Customer Key Center – International (CKCI) into the Certificate Authority Public Key field. You can click Expand View to expand this field if needed.

   **Note:** If the cryptogram contains spaces between blocks of hexadecimal characters, the spaces are automatically removed in the message sent to the security device.

2) Enter the SHA-1 hash value of the root Wincor Nixdorf public key into the SHA-1 Hash Value field.

3) Click Save.

**f)** If you are using a Thales e-Security HSM, you may skip to step i.

If you are using an Atalla NSP, the NSP validates the SHA-1 hash value of the root Wincor Nixdorf public key and generates a message digest of the root Wincor Nixdorf public key, which is displayed in the Message Digest field. Any validation errors are reported in screen messages on the Certificate Management window. If the SHA-1 hash value is incorrect, you must request Wincor Nixdorf to resend the root Wincor Nixdorf public key and SHA-1 hash value. The NSP uses command 126 to create a message digest value that is displayed in the Message Digest field on this tab.

<table>
<tr><th colspan="3">SHARED 912 SIGNATURES SETUP</th></tr>
<tr><td>g)</td><td>For an Atalla NSP only, encrypt the value returned in the Message Digest field from the previous step under variant 30 of the MFK using the SCT or SCA. Variant 30 is automatically assumed by both the SCT and SCA. Both the SCT and SCA require the message digest to be input as two 16-character fields. On the SCT, use the Utilities Menu -> Encrypt Challenge function to encrypt the message digest value. On the SCA, use the Calculate AKB/Cryptogram menu (Challenge/Response) to encrypt the message digest value. You must input the message digest as two 16-byte values (Left Challenge and Right Challenge on the SCT or Component block 1 and Component block 2 on the SCA). The results are two 16-byte values that comprise the root certificate hash value.</td><td></td></tr>
<tr><td>h)</td><td>Copy or enter the 32 byte (16 byte left and 16 byte right) values from the previous step into the Root Certificate Hash Value field on the Validate Certificates/Signatures tab.</td><td></td></tr>
<tr><td>i)</td><td>On the Validate Certificates/Signatures tab of the Certificate Management window perform the following:<br>1) Copy the root Wincor Nixdorf public key returned from the Wincor Nixdorf Customer Key Center – International (CKCI) into the Certificate Authority Public Key field on the Validate Certificates/Signatures tab of the Certificate Management window. You can click Expand View to expand this field if needed.<br>2) Copy the BASE24 system public key signature returned from the Wincor Nixdorf Customer Key Center – International (CKCI) into the Host Signature field on the Validate Certificates/Signatures tab of the Certificate Management window. You can click Expand View to expand this field if needed.<br>3) Click save.<br>The Atalla NSP validates the signature of the BASE24 system public key received from Wincor Nixdorf. If validated successfully, the BASE24 system public key signature is stored in the HSPKD.<br>For an Atalla NSP only, the root Wincor Nixdorf public key is validated using the hash value input from step h, imported, and stored in the HSPKD along with the message digest. Any validation errors are reported in screen messages on the Certificate Management window. If the key signature is incorrect, you must request that Wincor Nixdorf resend the BASE24 system public key signature.</td><td></td></tr>
<tr><td>j)</td><td>Once the above steps are completed, the following key data is entered into the database for the AKDS process:<br>• The BASE24 system’s public and private keys<br>• The signature of the BASE24 system’s public key signed by the root Wincor Nixdorf private key<br>• The root Wincor Nixdorf public key</td><td></td></tr>
<tr><th>9.</th><th colspan="2">DISABLE THE KEYGEN COMMAND (OPTIONAL)</th></tr>
<tr><td></td><td>After you have successfully generated an RSA key pair for the BASE24 system and have had the public key signed by the certificate authority, you may optionally disable use of the KEYGEN command.  This helps prevent the unintentional generation of a new RSA key pair, which, if it occurred, would require you to repeat the public key signing process. The KEYGEN command can be re-enabled if it is needed at a later time.</td><td></td></tr>
<tr><td>a)</td><td>Go to the Disable KEYGEN window of the ACI Desktop.</td><td></td></tr>
</table>

| SHARED 912 SIGNATURES SETUP | | |
|---|---|---|
| **b)** | Enter the value entered in the KEY SIGNER field on screen 22 of the ATD in the Host Name field.<br>Select **Wincor Signature** in the Device Type field.<br>Select **KEYGEN command is disabled** in the KEYGEN Command Status field. | |
| **c)** | Save the record. | |
| **End of Section** | | |

| SHARED 912 SIGNATURES TERMINAL DOWNLOADS | | |
|---|---|---|
| **1.** | **AUTHENTICATE AND EXCHANGE THE PUBLIC KEYS** | |
| | Before you can automatically download an ATM master key to the EPP of an ATM, the EPP and the BASE24 system must authenticate each other and exchange their public keys. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. (The DIEBOLD screen is used because the Diebold 10XX/478X device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD PUBLIC KEY field. | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the authentication and public key exchange. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^C1000, “9500LOADPUBLIC <terminal ID>” | |
| **2.** | **DOWNLOAD THE MASTER KEY** | |
| | Once the EPP has been successfully authenticated and has exchanged public keys with the BASE24 system, you can securely download the master key to the terminal. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. (The DIEBOLD screen is used because the Diebold 10XX/478X device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD MASTER KEY field | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the master key download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^C1000, “9500LOADMASTER <terminal ID>” | |
| **3.** | **DOWNLOAD THE MAC KEY AND/OR COMMUNICATIONS KEY** | |
| | Once the master key has been successfully downloaded to the terminal, you can securely download the MAC key and/or communications key, encrypted under the new master key. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | If you are downloading a new MAC key:<br><br>• Position the cursor in the LOAD MAC KEY field.<br><br>Press the F1 key. | |

| SHARED 912 SIGNATURES TERMINAL DOWNLOADS | | |
|---|---|---|
| **d)** | If you are downloading a new communications key:<br>• Enter a value of ‘2’ in the LOAD PIN KEY field.<br>• Press the F1 key while the cursor is positioned in the LOAD PIN KEY field. | |
| **e)** | Check the event log for the successful completion of the download. | |
| **4.** | **DOWNLOAD ALL KEYS (OPTIONAL)** | |
| | Optionally, you may load all of the aforementioned keys (public, master, MAC and communication keys) using a single DCT command, the LOAD ALL KEYS command. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. (The DIEBOLD screen is used because the Diebold 10XX/478X device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD ALL KEYS field. | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br>>DELIVER PROCESS P1A^C1000, “9500LOADALLKEYS <terminal ID>” | |
| **End of Section** | | |

# SHARED NDC+ SIGNATURES SETUP

## 1. APPLY SCNs

**a)** Apply the SCNs listed in the following file:

- Release 6.0 Version 6 $<VOL>.BA60UD09.SCNWNAK

If NCR ATMs are to be installed running Wincor's NDC+ Signature software, apply the SCNs listed in the following file:

- Release 6.0 Version 8 $<VOL>.BA60UD0B.SCNWAKD

## 2. MODIFY THE CUSTMACS FILE

**a)** The following Make flag has been added to the CUSTMACS file. Set the flag to TRUE to compile and bind the Shared AKDS extension to the NCR 5XXX device handler.

- atm_dh_n50_shrd_akd_on = $(TRUE)

## 3. MAKE THE SECURITY UTILITIES WITH THE AKDS EXTENSION

**a)** Restore the following AKDS Security Utilities extension subvolume and files.

| Subvolume.File | Type |
|---|---|
| BA60ASRC.BAASRCFM | Make |
| BA60ASRC.BAASRCM | Make |
| BA60ASRC.BAASRCMM | Make |
| BA60ASRC.SECAKDE | TAL External |
| BA60ASRC.SECAKDM | Make |
| BA60ASRC.SECAKDO | TAL object |
| BA60ASRC.SECAKDS | TAL source |

**b)** Run Make on the BA60SRC subvolume. The Make file for this subvolume is BA60SRC.BASRCM. Because the atm_dh_n50_shrd_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS Security Utilities extension will be compiled and bound into the main Security Utilities module.

## 4. MAKE THE NCR 5XXX DH WITH THE SHARED AKDS EXTENSION

**a)** Restore the following AKDS extension subvolume and files.

| Subvolume.File | Type |
|---|---|
| AT60AWN5.ATAW10FM | Make |
| AT60AWN5.ATAW10MM | Make |
| AT60AWN5.N50WAKDE | TAL External |
| AT60AWN5.N50WAKDG | TAL source |
| AT60AWN5.N50WAKDM | Make |
| AT60AWN5.N50WAKDO | TAL object |
| AT60AWN5.N50WAKDS | TAL source |

**b)** Run Make on the AT60N50 subvolume. The Make file for this subvolume is AT60N50.ATN50M. Because the atm_dh_n50_shrd_akd_on flag was set to TRUE in the CUSTMACS file, the Shared AKDS extension will be compiled and bound into the NCR 5XXX device handler.

| SHARED NDC+ SIGNATURES SETUP | | |
|---|---|---|
| **c)** | Replace the BASE24 system's current NCR 5XXX device handler object file with the newly compiled object file. | |
| **5.** | **ADD THE AKDS LCONF PARAMS** | |
| **a)** | Add the following params to the LCONF of each logical network that will support AKDS. | |
| **b)** | **ATM-DH-AKDS-TSS-TIMER**<br><br>The number of seconds the NCR 5XXX device handler process or Diebold 10XX/478X device handler process waits for a reply from the AKDS process before it considers the AKDS request to have timed out. In this case, the requested load is aborted. Valid values are 10–600 seconds. This param defaults to 600 seconds.<br><br>This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in hundredths of a second, to wait for a response from an HSM before the AKDS process considers the request to be timed out. The ATM-DH-AKDS-TSS-TIMER param value must be set to a value large enough to accommodate this length of time plus the time needed to send the request to and receive the response from the AKDS process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window. | |
| **c)** | **ATM-DH-1000-AKDS-SERIAL-NUM-VRFY**<br><br>An optional param indicating whether the NCR 5XXX device handler process verifies the EPP serial number received from the ATM when authenticating and exchanging public keys. If the value matches a valid EPP serial number stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:<br><br>Y = Yes, validate the EPP serial number<br>N = No, do not validate the EPP serial number<br><br>If this param is not configured or contains an invalid value, it defaults to a value of N.<br><br>**Note:** If the value of this param is set to Y, the serial numbers of the NCR or Wincor EPPs in the BASE24-atm system must be manually entered on the EPP Serial Number Configuration window before validation can be performed. | |
| **6.** | **CHECK THE BASE24-ATM TERMINAL DATA FILE (ATD) RECORDS** | |
| **a)** | For each NCR or Wincor terminal that supports AKDS, read its record on the BASE24-atm Terminal Data (ATD) file maintenance screen and go to screen 9. | |
| **b)** | The Master Key field should contain a key locator value that links this terminal's ATD record to a corresponding Channel Security Data (CSECD) record in the TSS database. If the Master Key field does not contain a key locator value, change it and update the record.<br><br>**Note:** The key locator value is the same as the ATD record's Term ID field (if the TSS database resides in a single logical network) or a two-byte LN-IND value followed by the Term ID (if the TSS database is partitioned across multiple logical networks). | |
| **c)** | Go to screen 1, and ensure that the NATIVE MESSAGE FORMAT field is set to a 1, to indicate that Wincor specification messages are being used. | |

| SHARED NDC+ SIGNATURES SETUP | | |
|---|---|---|
| **d)** | Go to screen 22, and set the KEY SIGNER field to a reasonable value. The field will default to 'HOST' if nothing is entered. | |
| **7.** | **CHECK THE CHANNEL SECURITY DATA (CSECD) RECORDS** | |
| **a)** | For each NCR or Wincor terminal that supports AKDS, read its record in the Channel Security Configuration window. Records are read according to the Channel ID field. The value of the Channel ID must equal the key locator value found in the terminal's BASE24-atm Terminal Data File (ATD) record, screen 9, Master Key field. | |
| **b)** | If a Channel Security Data record does not exist for this terminal, add it. | |
| **c)** | The Key Length of the Exchange Key should be set to **Double**.<br><br>The Key Length of the Inbound Key should be set to **Double**.<br><br>For NCR ATMs, the MAC option must be set to **Pseudo Triple DES Macing with 32 bits**. TSS 6.2 and above supports this for Thales and Eracom, Atalla has had this support all along.<br><br>Modify these fields if necessary, save the record, and warmboot all IS processes. | |
| **8.** | **GENERATE AN RSA KEY PAIR FOR THE BASE24 SYSTEM** | |
| **a)** | If you are using **NCR ATMs**, follow the directions in section 8 of the Native NCR 5XXX Setup Section.<br>**Note:** In the **Keygen** command**, HOST** should be replaced with the value entered in the KEY SIGNER field on screen 22 of the ATD, and the device type to be used is WINCORNCR instead of NCR.<br>If the customer already has an NCR public key established in their environment, they will need to copy that key and change the implementation type to Wincor NCR:<br><br>In DALCI:<br>Select * from host_public_key where pk_impl == 0,insertform;<br><br>Change the pk_impl to '3' in both records and copy them back into the file via dalci obey<br><br>If you are using **Wincor ATMs**, follow the directions here.<br><br>To start the process of submitting a public key and getting it signed by Wincor Nixdorf, contact your Wincor Nixdorf representative and request a Key Form.<br><br>**Note:** The procedures are essentially the same whether you are generating an RSA key pair for a test or production environment on the BASE24 system. For testing, a test EPP provided by Wincor Nixdorf must be used instead of a production EPP. When you initially contact Wincor Nixdorf to start the process of submitting a public key and getting it signed by Wincor Nixdorf, you must specify whether the key to be signed is for testing or production. You should use the same forms and procedures for getting both test and production keys. Wincor Nixdorf Customer Key Center – International (CKCI) signs test keys using a root public key that is different from the one it uses to sign production keys. | |
| **b)** | Generate an RSA public and private key pair for the BASE24 system. This is done one time only for a given BASE24 system unless the RSA key pair is corrupted in the TSS database.<br><br>To do so, go to the Process Control window of the ACI Desktop. | |

| SHARED NDC+ SIGNATURES SETUP | | |
|---|---|---|
| | • Select **XMLS** in the Destination field.<br>• Select **Deliver** in the Command field.<br>• Select **Transaction Security** in the Component field.<br>• Enter **KEYGEN HOST, WINCORSIG, , , , 2048, , DATE !** in the Command Text field.<br>**Note: HOST** should be replaced with the value entered in the KEY SIGNER field on screen 22 of the ATD.<br>**Note: DATE** should be replaced with the effective data required for the Public Key.<br>• Click on the green arrow in the upper left hand corner of the window to issue the command.<br>**Note:** The security device used to generate the RSA key pair is disabled during key generation, which may take up to 90 seconds. Ensure that the security device is PKI-enabled. | |
| **c)** | Access the Certificate Management window on the ACI Desktop to retrieve the BASE24 system public key certificate and format it in the manner expected by the Wincor Nixdorf Customer Key Center – International (CKCI). When the Certificate Management window is displayed, select the Host name, and select Wincor Signature as the device type in the Device Type field. Select the effective date required for the public key. The BASE24 system public key certificate is retrieved and displayed in the Host Public Key Data field. A 40-byte SHA-1 hash value is also generated and displayed in the SHA-1 Hash Value field. The BASE24 system public key is displayed in ASN.1 format. This is the message format required when sending the public key to CKCI. | |
| **d)** | Enter the 40-byte hash value displayed in the SHA-1 Hash Value field on the Host Public Key Data tab in the Digest (SHA-1) of Customer's Public Key field on the Wincor Nixdorf Key Form. Choose a Key Length of 2048 bits, and an Exchange Format of ASCII Hex. Choosing the ASCII Hex option ensures that the signature of your BASE24 system public key returned by Wincor Nixdorf will be in the correct format to enter into the ACI Desktop. After completing the remainder of the Key Form, fax it to the Wincor Nixdorf Customer Key Center – International (CKCI).<br><br>When requested by Wincor Nixdorf, copy the BASE24 system public key cryptogram from the Host Public Key Data field on the Certificate Management window into a text file. Send the text file on the appropriate media to Wincor Nixdorf following the format and procedures requested by the Wincor Nixdorf Customer Key Center – International (CKCI).<br><br>The Wincor Nixdorf Customer Key Center – International (CKCI) validates the SHA-1 hash value for your BASE24 system. Wincor Nixdorf will contact you to indicate whether the hash value is correct. If it is incorrect, you must resend the BASE24 system public key and hash value. If it is correct, the Wincor Nixdorf Customer Key Center – International (CKCI) creates a signature of the BASE24 system public key, generates a SHA-1 hash value for the root Wincor Nixdorf public key, and returns them back to you in the media format you requested. You will receive two files. Each file contains a key or hash value in ASCII Hex format. You must copy the contents of these files and paste them into fields on the Certificate Management window in subsequent steps as shown in the following table. | |

## SHARED NDC+ SIGNATURES SETUP

| Value (Wincor Nixdorf Filename) | Certificate Management Field |
|---|---|
| Root Wincor Nixdorf public key<br><br>pkSignWN<br><br>(pkSignWN<cust>T.mpk if this is a test key, or pkSignWN<cust>P.mpk if this is a production key) | Generate Message Digest > Certificate Authority Public Key (step e.1)<br><br>and<br><br>Validate Certificates/Signatures > Certificate Authority Public Key (step i.1) |
| SHA-1 hash of Wincor Nixdorf public key<br><br>Digest public key, calculated with SHA1<br><br>(pkSignWN<cust>T.mpk if this is for a test key, or pkSignWN<cust>P.mpk if this is for a production key) | Generate Message Digest > SHA-1 Hash Value (step e.2) |
| BASE24 system public key Signature<br><br>(siHSM<cust>T.sig if this is for a test key, or siHSM<cust>P.sig if this is for a production key) | Validate Certificates/Signatures >Host Signature (step i.2) |

**e)** On the Generate Message Digest tab of the Certificate Management window, perform the following:

1) Copy the root Wincor Nixdorf public key cryptogram from the Wincor Nixdorf Customer Key Center – International (CKCI) into the Certificate Authority Public Key field. You can click Expand View to expand this field if needed.

   **Note:** If the cryptogram contains spaces between blocks of hexadecimal characters, the spaces are automatically removed in the message sent to the security device.

2) Enter the SHA-1 hash value of the root Wincor Nixdorf public key into the SHA-1 Hash Value field.

3) Click Save.

**f)** If you are using a Thales e-Security HSM, you may skip to step i.

If you are using an Atalla NSP, the NSP validates the SHA-1 hash value of the root Wincor Nixdorf public key and generates a message digest of the root Wincor Nixdorf public key, which is displayed in the Message Digest field. Any validation errors are reported in screen messages on the Certificate Management window. If the SHA-1 hash value is incorrect, you must request Wincor

| SHARED NDC+ SIGNATURES SETUP | | |
|---|---|---|
| | Nixdorf to resend the root Wincor Nixdorf public key and SHA-1 hash value. The NSP uses command 126 to create a message digest value that is displayed in the Message Digest field on this tab. | |
| **g)** | For an Atalla NSP only, encrypt the value returned in the Message Digest field from the previous step under variant 30 of the MFK using the SCT or SCA. Variant 30 is automatically assumed by both the SCT and SCA. Both the SCT and SCA require the message digest to be input as two 16-character fields. On the SCT, use the Utilities Menu -> Encrypt Challenge function to encrypt the message digest value. On the SCA, use the Calculate AKB/Cryptogram menu (Challenge/Response) to encrypt the message digest value. You must input the message digest as two 16-byte values (Left Challenge and Right Challenge on the SCT or Component block 1 and Component block 2 on the SCA). The results are two 16-byte values that comprise the root certificate hash value. | |
| **h)** | Copy or enter the 32 byte (16 byte left and 16 byte right) values from the previous step into the Root Certificate Hash Value field on the Validate Certificates/Signatures tab. | |
| **i)** | On the Validate Certificates/Signatures tab of the Certificate Management window perform the following:<br>1) Copy the root Wincor Nixdorf public key returned from the Wincor Nixdorf Customer Key Center – International (CKCI) into the Certificate Authority Public Key field on the Validate Certificates/Signatures tab of the Certificate Management window. You can click Expand View to expand this field if needed.<br>2) Copy the BASE24 system public key signature returned from the Wincor Nixdorf Customer Key Center – International (CKCI) into the Host Signature field on the Validate Certificates/Signatures tab of the Certificate Management window. You can click Expand View to expand this field if needed.<br>3) Click save.<br><br>The Atalla NSP validates the signature of the BASE24 system public key received from Wincor Nixdorf. If validated successfully, the BASE24 system public key signature is stored in the HSPKD.<br><br>For an Atalla NSP only, the root Wincor Nixdorf public key is validated using the hash value input from step h, imported, and stored in the HSPKD along with the message digest. Any validation errors are reported in screen messages on the Certificate Management window. If the key signature is incorrect, you must request that Wincor Nixdorf resend the BASE24 system public key signature. | |
| **j)** | Once the above steps are completed, the following key data is entered into the database for the AKDS process:<br><br>• The BASE24 system's public and private keys<br>• The signature of the BASE24 system's public key signed by the root Wincor Nixdorf private key<br>• The root Wincor Nixdorf public key | |
| **9.** | **DISABLE THE KEYGEN COMMAND (OPTIONAL)** | |
| | After you have successfully generated an RSA key pair for the BASE24 system and have had the public key signed by the certificate authority, you may optionally disable use of the KEYGEN command. This helps prevent the unintentional generation of a new RSA | |

| SHARED NDC+ SIGNATURES SETUP | | |
|---|---|---|
| | key pair, which, if it occurred, would require you to repeat the public key signing process. The KEYGEN command can be re-enabled if it is needed at a later time. | |
| **a)** | Go to the Disable KEYGEN window of the ACI Desktop. | |
| **b)** | Enter the value entered in the KEY SIGNER field on screen 22 of the ATD in the Host Name field.<br>Select **Wincor Signature** in the Device Type field.<br>Select **KEYGEN command is disabled** in the KEYGEN Command Status field. | |
| **c)** | Save the record. | |
| **End of Section** | | |

| | | |
|---|---|---|
| **SHARED NDC+ SIGNATURES TERMINAL DOWNLOADS** | | |
| **1.** | **AUTHENTICATE AND EXCHANGE THE PUBLIC KEYS** | |
| | Before you can automatically download an ATM master key to the EPP of an ATM, the EPP and the BASE24 system must authenticate each other and exchange their public keys. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The NCR 5XXX LOAD COMMANDS screen will appear. (The NCR screen is used because the NCR 5XXX device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD PUBLIC KEY field. | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the authentication and public key exchange. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, “9500LOADPUBLIC <terminal ID>” | |
| **2.** | **DOWNLOAD THE MASTER KEY** | |
| | Once the EPP has been successfully authenticated and has exchanged public keys with the BASE24 system, you can securely download the master key to the terminal. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The NCR 5XXX LOAD COMMANDS screen will appear. (The NCR screen is used because the NCR 5XXX device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD MASTER KEY field | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the master key download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^ N50, “9500LOADMASTER <terminal ID>” | |
| **3.** | **DOWNLOAD THE MAC KEY AND/OR COMMUNICATIONS KEY** | |
| | Once the master key has been successfully downloaded to the terminal, you can securely download the MAC key and/or communications key, encrypted under the new master key. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | If you are downloading a new MAC key:<br><br>• Position the cursor in the LOAD MAC KEY field.<br><br>Press the F1 key. | |

| SHARED NDC+ SIGNATURES TERMINAL DOWNLOADS | | |
|---|---|---|
| **d)** | If you are downloading a new communications key:<br>• Enter a value of ‘2’ in the LOAD PIN KEY field.<br>• Press the F1 key while the cursor is positioned in the LOAD PIN KEY field. | |
| **e)** | Check the event log for the successful completion of the download. | |
| **4.** | **DOWNLOAD ALL KEYS (OPTIONAL)** | |
| | Optionally, you may load all of the aforementioned keys (public, master, MAC and communication keys) using a single DCT command, the LOAD ALL KEYS command. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The NCR 5XXX LOAD COMMANDS screen will appear. (The NCR screen is used because the NCR 5XXX device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD ALL KEYS field. | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br>>DELIVER PROCESS P1A^ N50, “9500LOADALLKEYS <terminal ID>” | |
| **End of Section** | | |

| SHARED 912 CERTIFICATES SETUP | | |
|---|---|---|
| **1.** | **APPLY SCNs** | |
| **a)** | Apply the SCNs listed in the following file:<br><br>• Release 6.0 Version 7 $<VOL>.BA60UD0A.SCNWDCT | |
| **2.** | **MODIFY THE CUSTMACS FILE** | |
| **a)** | The following Make flag already exists in the CUSTMACS file. Ensure the flag is set to TRUE to compile and bind the Shared AKDS extension to the Diebold 10XX/478X device handler.<br><br>• atm_dh_c1000_shrd_akd_on = $(TRUE) | |
| **3.** | **MAKE THE SECURITY UTILITIES WITH THE AKDS EXTENSION** | |
| **a)** | Restore the following AKDS Security Utilities extension subvolume and files.<br><br><u>Subvolume.File</u> <u>Type</u><br>BA60ASRC.BAASRCFM Make<br>BA60ASRC.BAASRCM Make<br>BA60ASRC.BAASRCMM Make<br>BA60ASRC.SECAKDE TAL External<br>BA60ASRC.SECAKDM Make<br>BA60ASRC.SECAKDO TAL object<br>BA60ASRC.SECAKDS TAL source | |
| **b)** | Run Make on the BA60SRC subvolume. The Make file for this subvolume is BA60SRC.BASRCM. Because the atm_dh_c1000_shrd_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS Security Utilities extension will be compiled and bound into the main Security Utilities module. | |
| **4.** | **MAKE THE DIEBOLD 10XX/478X DH WITH THE SHARED AKDS EXTENSION** | |
| **a)** | Restore the following AKDS extension subvolume and files.<br><br><u>Subvolume.File</u> <u>Type</u><br>AT60AW10.ATAW10FM Make<br>AT60AW10.ATAW10MM Make<br>AT60AW10.C10WAKDE TAL External<br>AT60AW10.C10WAKDG TAL source<br>AT60AW10.C10WAKDM Make<br>AT60AW10.C10WAKDO TAL object<br>AT60AW10.C10WAKDS TAL source | |
| **b)** | Run Make on the AT601000 subvolume. The Make file for this subvolume is AT601000.AT1000M. Because the atm_dh_c1000_shrd_akd_on flag was set to TRUE in the CUSTMACS file, the Shared AKDS extension will be compiled and bound into the Diebold 10XX/478X device handler. | |
| **c)** | Replace the BASE24 system’s current Diebold 10XX/478X device handler object file with the newly compiled object file. | |

| SHARED 912 CERTIFICATES SETUP | | |
|---|---|---|
| **5.** | **ADD THE AKDS LCONF PARAMS** | |
| **a)** | Ensure that the following params exist in the LCONF of each logical network that will support AKDS. | |
| **b)** | **ATM-DH-AKDS-TSS-TIMER**<br><br>The number of seconds the NCR 5XXX device handler process or Diebold 10XX/478X device handler process waits for a reply from the AKDS process before it considers the AKDS request to have timed out. In this case, the requested load is aborted. Valid values are 10–600 seconds. This param defaults to 600 seconds.<br><br>This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in hundredths of a second, to wait for a response from an HSM before the AKDS process considers the request to be timed out. The ATM-DH-AKDS-TSS-TIMER param value must be set to a value large enough to accommodate this length of time plus the time needed to send the request to and receive the response from the AKDS process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window. | |
| **c)** | **ATM-DH-1000-AKDS-SERIAL-NUM-VRFY**<br><br>An optional param indicating whether the Diebold 10XX/478X device handler process verifies the EPP serial number received from the ATM when authenticating and exchanging public keys. If the value matches a valid EPP serial number stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:<br><br>Y = Yes, validate the EPP serial number<br>N = No, do not validate the EPP serial number<br><br>If this param is not configured or contains an invalid value, it defaults to a value of N.<br><br>**Note:** If the value of this param is set to Y, the serial numbers of the NCR or Wincor EPPs in the BASE24-atm system must be manually entered on the EPP Serial Number Configuration window before validation can be performed. | |
| **6.** | **CHECK THE BASE24-ATM TERMINAL DATA FILE (ATD) RECORDS** | |
| **a)** | For each Diebold terminal that supports AKDS, read its record on the BASE24-atm Terminal Data (ATD) file maintenance screen and go to screen 9. | |
| **b)** | The Master Key field should contain a key locator value that links this terminal’s ATD record to a corresponding Channel Security Data (CSECD) record in the TSS database. If the Master Key field does not contain a key locator value, change it and update the record.<br><br>**Note:** The key locator value is the same as the ATD record’s Term ID field (if the TSS database resides in a single logical network) or a two-byte LN-IND value followed by the Term ID (if the TSS database is partitioned across multiple logical networks). | |
| **c)** | Go to screen 1, and ensure that the NATIVE MESSAGE FORMAT field is set to a 1, to indicate that Wincor specification messages are being used. | |
| **d)** | Go to screen 22, and set the KEY SIGNER field to a reasonable value. The field will default to ‘HOST’ if nothing is entered. | |

| SHARED 912 CERTIFICATES SETUP | | |
|---|---|---|
| **7.** | **CHECK THE CHANNEL SECURITY DATA (CSECD) RECORDS** | |
| **a)** | For each Diebold terminal that supports AKDS, read its record in the Channel Security Configuration window. Records are read according to the Channel ID field. The value of the Channel ID must equal the key locator value found in the terminal's BASE24-atm Terminal Data File (ATD) record, screen 9, Master Key field. | |
| **b)** | If a Channel Security Data record does not exist for this terminal, add it. | |
| **c)** | The Key Length of the Exchange Key should be set to **Double**.<br><br>The Key Length of the Inbound Key should be set to **Double**.<br><br>Modify these fields if necessary, save the record, and warmboot all IS processes. | |
| **8.** | **GENERATE AN RSA KEY PAIR FOR THE BASE24 SYSTEM** | |
| **a)** | As you are using Diebold ATMs for this solution, follow the directions in section 8 of the Native Diebold 10XX/478X Setup Section.<br><br>**Note:** In the **Keygen** command**, HOST** should be replaced with the value entered in the KEY SIGNER field on screen 22 of the ATD. | |
| **9.** | **DISABLE THE KEYGEN COMMAND (OPTIONAL)** | |
| | After you have successfully generated an RSA key pair for the BASE24 system and have had the public key signed by the certificate authority, you may optionally disable use of the KEYGEN command. This helps prevent the unintentional generation of a new RSA key pair, which, if it occurred, would require you to repeat the public key signing process. The KEYGEN command can be re-enabled if it is needed at a later time. | |
| **a)** | Go to the Disable KEYGEN window of the ACI Desktop. | |
| **b)** | Enter the value entered in the KEY SIGNER field on screen 22 of the ATD in the Host Name field.<br><br>Select **Diebold** in the Device Type field.<br><br>Select **KEYGEN command is disabled** in the KEYGEN Command Status field. | |
| **c)** | Save the record. | |
| **End of Section** | | |

| SHARED 912 CERTIFICATES TERMINAL DOWNLOADS | | |
|---|---|---|
| **1.** | **AUTHENTICATE AND EXCHANGE THE PUBLIC KEYS** | |
| | Before you can automatically download an ATM master key to the EPP of an ATM, the EPP and the BASE24 system must authenticate each other and exchange their public keys. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. (The DIEBOLD screen is used because the Diebold 10XX/478X device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD PUBLIC KEY field. | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the authentication and public key exchange. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^C1000, “9500LOADPUBLIC <terminal ID>” | |
| **2.** | **DOWNLOAD THE MASTER KEY** | |
| | Once the EPP has been successfully authenticated and has exchanged public keys with the BASE24 system, you can securely download the master key to the terminal. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. (The DIEBOLD screen is used because the Diebold 10XX/478X device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD MASTER KEY field | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the master key download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^C1000, “9500LOADMASTER <terminal ID>” | |
| **3.** | **DOWNLOAD THE MAC KEY AND/OR COMMUNICATIONS KEY** | |
| | Once the master key has been successfully downloaded to the terminal, you can securely download the MAC key and/or communications key, encrypted under the new master key. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | If you are downloading a new MAC key:<br><br>• Position the cursor in the LOAD MAC KEY field.<br><br>Press the F1 key. | |

| SHARED 912 CERTIFICATES TERMINAL DOWNLOADS | | |
|---|---|---|
| **d)** | If you are downloading a new communications key:<br>• Enter a value of ‘2’ in the LOAD PIN KEY field.<br>• Press the F1 key while the cursor is positioned in the LOAD PIN KEY field. | |
| **e)** | Check the event log for the successful completion of the download. | |
| **4.** | **DOWNLOAD ALL KEYS (OPTIONAL)** | |
| | Optionally, you may load all of the aforementioned keys (public, master, MAC and communication keys) using a single DCT command, the LOAD ALL KEYS command. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The DIEBOLD 10XX COMMANDS screen will appear. (The DIEBOLD screen is used because the Diebold 10XX/478X device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD ALL KEYS field. | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br>>DELIVER PROCESS P1A^C1000, “9500LOADALLKEYS <terminal ID>” | |
| **End of Section** | | |

# SHARED NDC+ CERTIFICATES SETUP

## 1. APPLY SCNs

**a)** Apply the SCNs listed in the following file:

- Release 6.0 Version 7 $<VOL>.BA60UD0A.SCNWNCT

## 2. MODIFY THE CUSTMACS FILE

**a)** The following Make flag already exists in the CUSTMACS file. Ensure the flag is set to TRUE to compile and bind the Shared AKDS extension to the NCR 5XXX device handler.

- atm_dh_n50_shrd_akd_on = $(TRUE)

## 3. MAKE THE SECURITY UTILITIES WITH THE AKDS EXTENSION

**a)** Restore the following AKDS Security Utilities extension subvolume and files.

| Subvolume.File | Type |
|---|---|
| BA60ASRC.BAASRCFM | Make |
| BA60ASRC.BAASRCM | Make |
| BA60ASRC.BAASRCMM | Make |
| BA60ASRC.SECAKDE | TAL External |
| BA60ASRC.SECAKDM | Make |
| BA60ASRC.SECAKDO | TAL object |
| BA60ASRC.SECAKDS | TAL source |

**b)** Run Make on the BA60SRC subvolume. The Make file for this subvolume is BA60SRC.BASRCM. Because the atm_dh_n50_shrd_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS Security Utilities extension will be compiled and bound into the main Security Utilities module.

## 4. MAKE THE NCR 5XXX DH WITH THE SHARED AKDS EXTENSION

**a)** Restore the following AKDS extension subvolume and files.

| Subvolume.File | Type |
|---|---|
| AT60AWN5.ATAW10FM | Make |
| AT60AWN5.ATAW10MM | Make |
| AT60AWN5.N50WAKDE | TAL External |
| AT60AWN5.N50WAKDG | TAL source |
| AT60AWN5.N50WAKDM | Make |
| AT60AWN5.N50WAKDO | TAL object |
| AT60AWN5.N50WAKDS | TAL source |

**b)** Run Make on the AT60N50 subvolume. The Make file for this subvolume is AT60N50.ATN50M. Because the atm_dh_n50_shrd_akd_on flag was set to TRUE in the CUSTMACS file, the Shared AKDS extension will be compiled and bound into the NCR 5XXX device handler.

**c)** Replace the BASE24 system's current NCR 5XXX device handler object file with the newly compiled object file.

| SHARED NDC+ CERTIFICATES SETUP | | |
|---|---|---|
| **5.** | **ADD THE AKDS LCONF PARAMS** | |
| **a)** | Ensure that the following params exist in the LCONF of each logical network that will support AKDS. | |
| **b)** | **ATM-DH-AKDS-TSS-TIMER**<br><br>The number of seconds the NCR 5XXX device handler process or Diebold 10XX/478X device handler process waits for a reply from the AKDS process before it considers the AKDS request to have timed out. In this case, the requested load is aborted. Valid values are 10–600 seconds. This param defaults to 600 seconds.<br><br>This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in hundredths of a second, to wait for a response from an HSM before the AKDS process considers the request to be timed out. The ATM-DH-AKDS-TSS-TIMER param value must be set to a value large enough to accommodate this length of time plus the time needed to send the request to and receive the response from the AKDS process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window. | |
| **c)** | **ATM-DH-1000-AKDS-SERIAL-NUM-VRFY**<br><br>An optional param indicating whether the NCR 5XXX device handler process verifies the EPP serial number received from the ATM when authenticating and exchanging public keys. If the value matches a valid EPP serial number stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:<br><br>Y = Yes, validate the EPP serial number<br>N = No, do not validate the EPP serial number<br><br>If this param is not configured or contains an invalid value, it defaults to a value of N.<br><br>**Note:** If the value of this param is set to Y, the serial numbers of the NCR or Wincor EPPs in the BASE24-atm system must be manually entered on the EPP Serial Number Configuration window before validation can be performed. | |
| **6.** | **CHECK THE BASE24-ATM TERMINAL DATA FILE (ATD) RECORDS** | |
| **a)** | For each Diebold terminal that supports AKDS, read its record on the BASE24-atm Terminal Data (ATD) file maintenance screen and go to screen 9. | |
| **b)** | The Master Key field should contain a key locator value that links this terminal's ATD record to a corresponding Channel Security Data (CSECD) record in the TSS database. If the Master Key field does not contain a key locator value, change it and update the record.<br><br>**Note:** The key locator value is the same as the ATD record's Term ID field (if the TSS database resides in a single logical network) or a two-byte LN-IND value followed by the Term ID (if the TSS database is partitioned across multiple logical networks). | |
| **c)** | Go to screen 1, and ensure that the NATIVE MESSAGE FORMAT field is set to a 1, to indicate that Wincor specification messages are being used. | |
| **d)** | Go to screen 22, and set the KEY SIGNER field to a reasonable value. The field will default to 'HOST' if nothing is entered. | |

| SHARED NDC+ CERTIFICATES SETUP | | |
|---|---|---|
| **7.** | **CHECK THE CHANNEL SECURITY DATA (CSECD) RECORDS** | |
| **a)** | For each Diebold terminal that supports AKDS, read its record in the Channel Security Configuration window. Records are read according to the Channel ID field. The value of the Channel ID must equal the key locator value found in the terminal's BASE24-atm Terminal Data File (ATD) record, screen 9, Master Key field. | |
| **b)** | If a Channel Security Data record does not exist for this terminal, add it. | |
| **c)** | The Key Length of the Exchange Key should be set to **Double**.<br><br>The Key Length of the Inbound Key should be set to **Double**.<br><br>Modify these fields if necessary, save the record, and warmboot all IS processes. | |
| **8.** | **GENERATE AN RSA KEY PAIR FOR THE BASE24 SYSTEM** | |
| **a)** | As you are using Diebold ATMs for this solution, follow the directions in section 8 of the Native Diebold 10XX/478X Setup Section.<br><br>**Note:** In the **Keygen** command**, HOST** should be replaced with the value entered in the KEY SIGNER field on screen 22 of the ATD. | |
| **9.** | **DISABLE THE KEYGEN COMMAND (OPTIONAL)** | |
| | After you have successfully generated an RSA key pair for the BASE24 system and have had the public key signed by the certificate authority, you may optionally disable use of the KEYGEN command. This helps prevent the unintentional generation of a new RSA key pair, which, if it occurred, would require you to repeat the public key signing process. The KEYGEN command can be re-enabled if it is needed at a later time. | |
| **a)** | Go to the Disable KEYGEN window of the ACI Desktop. | |
| **b)** | Enter the value entered in the KEY SIGNER field on screen 22 of the ATD in the Host Name field.<br><br>Select **Diebold** in the Device Type field.<br><br>Select **KEYGEN command is disabled** in the KEYGEN Command Status field. | |
| **c)** | Save the record. | |
| **End of Section** | | |

| SHARED NDC+ CERTIFICATES TERMINAL DOWNLOADS | | |
|---|---|---|
| **1.** | **AUTHENTICATE AND EXCHANGE THE PUBLIC KEYS** | |
| | Before you can automatically download an ATM master key to the EPP of an ATM, the EPP and the BASE24 system must authenticate each other and exchange their public keys. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The NCR 5XXX LOAD COMMANDS screen will appear. (The NCR screen is used because the NCR 5XXX device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD PUBLIC KEY field. | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the authentication and public key exchange. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^N50, “9500LOADPUBLIC <terminal ID>” | |
| **2.** | **DOWNLOAD THE MASTER KEY** | |
| | Once the EPP has been successfully authenticated and has exchanged public keys with the BASE24 system, you can securely download the master key to the terminal. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The NCR 5XXX LOAD COMMANDS screen will appear. (The NCR screen is used because the NCR 5XXX device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD MASTER KEY field | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the master key download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br><br>>DELIVER PROCESS P1A^ N50, “9500LOADMASTER <terminal ID>” | |
| **3.** | **DOWNLOAD THE MAC KEY AND/OR COMMUNICATIONS KEY** | |
| | Once the master key has been successfully downloaded to the terminal, you can securely download the MAC key and/or communications key, encrypted under the new master key. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The NCR 5XXX LOAD COMMANDS screen will appear. | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | If you are downloading a new MAC key:<br><br>• Position the cursor in the LOAD MAC KEY field.<br><br>Press the F1 key. | |

| SHARED NDC+ CERTIFICATES TERMINAL DOWNLOADS | | |
|---|---|---|
| **d)** | If you are downloading a new communications key:<br>• Enter a value of '2' in the LOAD PIN KEY field.<br>• Press the F1 key while the cursor is positioned in the LOAD PIN KEY field. | |
| **e)** | Check the event log for the successful completion of the download. | |
| **4.** | **DOWNLOAD ALL KEYS (OPTIONAL)** | |
| | Optionally, you may load all of the aforementioned keys (public, master, MAC and communication keys) using a single DCT command, the LOAD ALL KEYS command. | |
| **a)** | Go to Device Control Terminal (DCT) screen 40. The NCR 5XXX LOAD COMMANDS screen will appear. (The NCR screen is used because the NCR 5XXX device handler is being used). | |
| **b)** | Enter the ATM TERMINAL ID. | |
| **c)** | Position the cursor in the LOAD ALL KEYS field. | |
| **d)** | Press the F1 key. | |
| **e)** | Check the event log for the successful completion of the download. | |
| | Alternatively, you may enter the following command from the NCS screen or NCPCOM:<br>>DELIVER PROCESS P1A^ N50, "9500LOADALLKEYS <terminal ID>" | |
| **End of Section** | | |

## NATIVE TIDEL SETUP

| | | |
|---|---|---|
| **1.** | **APPLY SCNs** | |
| **a)** | Apply the SCNs listed in the following files:<br><br>• Release 6.0 Version 9 $<VOL>.BA60UD0C.SCNTIAK | |
| **2.** | **MODIFY THE CUSTMACS FILE** | |
| **a)** | The following Make flag has been added to the CUSTMACS file. Set the flag to TRUE to compile and bind the AKDS extension to the Tidel device handler.<br><br>• atm_dh_tidel_akd_on = $(TRUE) | |
| **3.** | **MAKE THE SECURITY UTILITIES WITH THE AKDS EXTENSION** | |
| **a)** | Restore the following AKDS Security Utilities extension subvolume and files.<br><br><u>Subvolume.File</u> — <u>Type</u><br>BA60ASRC.BAASRCFM — Make<br>BA60ASRC.BAASRCM — Make<br>BA60ASRC.BAASRCMM — Make<br>BA60ASRC.SECAKDE — TAL External<br>BA60ASRC.SECAKDM — Make<br>BA60ASRC.SECAKDO — TAL object<br>BA60ASRC.SECAKDS — TAL source | |
| **b)** | Run Make on the BA60SRC subvolume. The Make file for this subvolume is BA60SRC.BASRCM. Because the atm_dh_tidel_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS Security Utilities extension will be compiled and bound into the main Security Utilities module. | |
| **4.** | **MAKE THE TIDEL DH WITH THE AKDS EXTENSION** | |
| **a)** | Restore the following AKDS extension subvolume and files.<br><br><u>Subvolume.File</u> — <u>Type</u><br>AT60ATDL.ATATDLFM — Make<br>AT60ATDL.ATATDLMM — Make<br>AT60ATDL.TIDLAKDE — TAL External<br>AT60ATDL.TIDLAKDG — TAL source<br>AT60ATDL.TIDLAKDM — Make<br>AT60ATDL.TIDLAKDO — TAL object<br>AT60ATDL.TIDLAKDS — TAL source | |
| **b)** | Run Make on the AT60TIDL subvolume. The Make file for this subvolume is AT60TIDL.ATTIDLM. Because the atm_dh_tidel_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS extension will be compiled and bound into the Tidel device handler. | |
| **c)** | Replace the BASE24 system's current Tidel device handler object file with the newly compiled object file. | |

| NATIVE TIDEL SETUP | | |
|---|---|---|
| **5.** | **ADD THE AKDS LCONF PARAMS** | |
| **a)** | Add the following params to the LCONF of each logical network that will support AKDS. | |
| **b)** | **ATM-DH-AKDS-TSS-TIMER**<br><br>The number of seconds the NCR 5XXX device handler process, Diebold 10XX/478X device handler process, Tidel device handler process or Triton device handler process waits for a reply from the AKDS process before it considers the AKDS request to have timed out. In this case, the requested load is aborted. Valid values are 10–600 seconds. This param defaults to 600 seconds.<br><br>This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in hundredths of a second, to wait for a response from an HSM before the AKDS process considers the request to be timed out. The ATM-DH-AKDS-TSS-TIMER param value must be set to a value large enough to accommodate this length of time plus the time needed to send the request to and receive the response from the AKDS process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window. | |
| **c)** | **ATM-DH-TIDL-AKDS-SERIAL-NUM-VRFY**<br><br>An optional param indicating whether the Tidel device handler process verifies the EPP serial number received from the ATM when authenticating and exchanging public keys. If the value matches a valid EPP serial number stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:<br><br>Y = Yes, validate the EPP serial number<br>N = No, do not validate the EPP serial number<br><br>If this param is not configured or contains an invalid value, it defaults to a value of N.<br><br>**Note:** If the value of this param is set to Y, the serial numbers of the NCR EPPs in the BASE24-atm system must be manually entered on the EPP Serial Number Configuration window before validation can be performed. | |
| **d)** | **ATM-DH-TIDL-AKDS-HOST-LOAD-TIMER**<br>An optional param indicating the number of seconds that the ATM waits for a HOST RKM command message to be sent during an AKDS load, before aborting the load. Valid values are 10–600 seconds. This param defaults to 60 seconds. | |
| **6.** | **CHECK THE BASE24-ATM TERMINAL DEFINITION FILE (TDF) RECORDS** | |
| **a)** | For each Tidel terminal that supports AKDS, read its record on the BASE24-atm Terminal Definition (ATD) file maintenance screen and go to screen 9. | |
| **b)** | The Master Key field should contain a key locator value that links this terminal's TDF record to a corresponding Channel Security Data (CSECD) record in the TSS database. If the Master Key field does not contain a key locator value, change it and update the record.<br><br>**Note:** The key locator value is the same as the TDF record's Term ID field (if the TSS database resides in a single logical network) or a two-byte LN-IND value followed by the | |

| NATIVE TIDEL SETUP | | |
|---|---|---|
| | Term ID (if the TSS database is partitioned across multiple logical networks). | |
| **c)** | Go to screen 1, and ensure that the NATIVE MESSAGE FORMAT field is set to a 0, to indicate that native Tidel messages are being used. | |
| **d)** | Go to screen 22, and set the KEY SIGNER field to a reasonable value. The field will default to 'HOST' if nothing is entered. | |
| **7.** | **CHECK THE CHANNEL SECURITY DATA (CSECD) RECORDS** | |
| **a)** | For each Tidel terminal that supports AKDS, read its record in the Channel Security Configuration window. Records are read according to the Channel ID field. The value of the Channel ID must equal the key locator value found in the terminal's BASE24-atm Terminal Definition File (TDF) record, screen 9, Master Key field. | |
| **b)** | If a Channel Security Data record does not exist for this terminal, add it. | |
| **c)** | The Key Length of the Exchange Key should be set to **Double**.<br><br>The Key Length of the Inbound Key should be set to **Double**.<br><br>The MAC option must be set to **Pseudo Triple DES Macing with 32 bits**. TSS 6.2 and above supports this for Thales and Eracom, Atalla has had this support all along.<br><br>Modify these fields if necessary, save the record, and warmboot all IS processes. | |
| **8.** | **GENERATE AN RSA KEY PAIR FOR THE BASE24 SYSTEM** | |
| **a)** | Tidel is now owned by NCR, so follow the directions in section 8 of the Native NCR 5XXX Basic Setup Section. | |
| **9.** | **DISABLE THE KEYGEN COMMAND (OPTIONAL)** | |
| | After you have successfully generated an RSA key pair for the BASE24 system and have had the public key signed by the certificate authority, you may optionally disable use of the KEYGEN command. This helps prevent the unintentional generation of a new RSA key pair, which, if it occurred, would require you to repeat the public key signing process. The KEYGEN command can be re-enabled if it is needed at a later time. | |
| **a)** | Go to the Disable KEYGEN window of the ACI Desktop. | |
| **b)** | Enter the value entered in the KEY SIGNER field on screen 22 of the ATD in the Host Name field<br><br>Select **NCR** in the Device Type field.<br><br>Select **KEYGEN command is disabled** in the KEYGEN Command Status field. | |
| **c)** | Save the record. | |
| **End of Section** | | |

| TIDEL TERMINAL DOWNLOADS | | |
|---|---|---|
| **1.** | **DOWNLOAD ALL KEYS** | |
| | You may load all of the keys (public, master, MAC and communication keys) using a single NCPCOM command, the LOAD ALL KEYS command. | |
| **a)** | Go to NCPCOM | |
| **b)** | Enter DELIVER PROCESS <process-name>, "9500LOADALLKEYS <terminal-id>" | |
| **c)** | After the next transaction is received by the device handler, an AKDS load will be initiated. | |
| **d)** | Check the event log for the successful completion of the download. | |
| **End of Section** | | |

<table>
<tr><th colspan="3">NATIVE TRITON SETUP</th></tr>
<tr><td>1.</td><td colspan="2">APPLY SCNs</td></tr>
<tr><td>a)</td><td>Apply the SCNs listed in the following files:<br><br>• Release 6.0 Version 9 $<VOL>.BA60UD0C.SCNTRAK</td><td></td></tr>
<tr><td>2.</td><td colspan="2">MODIFY THE CUSTMACS FILE</td></tr>
<tr><td>a)</td><td>The following Make flag has been added to the CUSTMACS file. Set the flag to TRUE to compile and bind the AKDS extension to the Triton device handler.<br><br>• atm_dh_triton_akd_on = $(TRUE)</td><td></td></tr>
<tr><td>3.</td><td colspan="2">MAKE THE SECURITY UTILITIES WITH THE AKDS EXTENSION</td></tr>
<tr><td>a)</td><td>Restore the following AKDS Security Utilities extension subvolume and files.<br><br><u>Subvolume.File</u> <u>Type</u><br>BA60ASRC.BAASRCFM Make<br>BA60ASRC.BAASRCM Make<br>BA60ASRC.BAASRCMM Make<br>BA60ASRC.SECAKDE TAL External<br>BA60ASRC.SECAKDM Make<br>BA60ASRC.SECAKDO TAL object<br>BA60ASRC.SECAKDS TAL source</td><td></td></tr>
<tr><td>b)</td><td>Run Make on the BA60SRC subvolume. The Make file for this subvolume is BA60SRC.BASRCM. Because the atm_dh_triton_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS Security Utilities extension will be compiled and bound into the main Security Utilities module.</td><td></td></tr>
<tr><td>4.</td><td colspan="2">MAKE THE TRITON DH WITH THE AKDS EXTENSION</td></tr>
<tr><td>a)</td><td>Restore the following AKDS extension subvolume and files.<br><br><u>Subvolume.File</u> <u>Type</u><br>AT60ATRT.ATATRTFM Make<br>AT60ATRT.ATATRTMM Make<br>AT60ATRT.ATATRTM Make<br>AT60ATRT.TRITAKDE TAL External<br>AT60ATRT.TRITAKDG TAL source<br>AT60ATRT.TRITAKDM Make<br>AT60ATRT.TRITAKDO TAL object<br>AT60ATRT.TRITAKDS TAL source</td><td></td></tr>
<tr><td>b)</td><td>Run Make on the AT60TRIT subvolume. The Make file for this subvolume is AT60TRITL.ATTRITM. Because the atm_dh_triton_akd_on flag was set to TRUE in the CUSTMACS file, the AKDS extension will be compiled and bound into the Triton device handler.</td><td></td></tr>
<tr><td>c)</td><td>Replace the BASE24 system’s current Triton device handler object file with the newly compiled object file.</td><td></td></tr>
</table>

| NATIVE TRITON SETUP | | |
|---|---|---|
| **5.** | **ADD THE AKDS LCONF PARAMS** | |
| **a)** | Add the following params to the LCONF of each logical network that will support AKDS. | |
| **b)** | **ATM-DH-AKDS-TSS-TIMER**<br><br>The number of seconds the NCR 5XXX device handler process, Diebold 10XX/478X device handler process, Tidel device handler process or Triton device handler process waits for a reply from the AKDS process before it considers the AKDS request to have timed out. In this case, the requested load is aborted. Valid values are 10–600 seconds. This param defaults to 600 seconds.<br><br>This param must be set in consideration of the Timeout Limit field value on the Security Device Configuration window. The Timeout Limit field value specifies the maximum time, in hundredths of a second, to wait for a response from an HSM before the AKDS process considers the request to be timed out. The ATM-DH-AKDS-TSS-TIMER param value must be set to a value large enough to accommodate this length of time plus the time needed to send the request to and receive the response from the AKDS process. Thus, it should always be set to a value greater than that set in the Timeout Limit field on the Security Device Configuration window. | |
| **c)** | **ATM-DH-TRIT-AKDS-SERIAL-NUM-VRFY**<br><br>An optional param indicating whether the Triton device handler process verifies the EPP serial number received from the ATM when authenticating and exchanging public keys. If the value matches a valid EPP serial number stored in the EPP Serial Number File (EPNUMD), it is considered to be valid. Valid values are as follows:<br><br>Y = Yes, validate the EPP serial number<br>N = No, do not validate the EPP serial number<br><br>If this param is not configured or contains an invalid value, it defaults to a value of N.<br><br>**Note:** If the value of this param is set to Y, the serial numbers of the NCR EPPs in the BASE24-atm system must be manually entered on the EPP Serial Number Configuration window before validation can be performed. | |
| **6.** | **CHECK THE BASE24-ATM TERMINAL DEFINITION FILE (TDF) RECORDS** | |
| **a)** | For each Triton terminal that supports AKDS, read its record on the BASE24-atm Terminal Definition (ATD) file maintenance screen and go to screen 9. | |
| **b)** | The Master Key field should contain a key locator value that links this terminal's TDF record to a corresponding Channel Security Data (CSECD) record in the TSS database. If the Master Key field does not contain a key locator value, change it and update the record.<br><br>**Note:** The key locator value is the same as the TDF record's Term ID field (if the TSS database resides in a single logical network) or a two-byte LN-IND value followed by the Term ID (if the TSS database is partitioned across multiple logical networks). | |

| NATIVE TRITON SETUP | | |
|---|---|---|
| **c)** | Go to screen 1, and ensure that the NATIVE MESSAGE FORMAT field is set to a 0, to indicate that native Triton messages are being used. | |
| **d)** | Go to screen 22, and set the KEY SIGNER field to a reasonable value. The field will default to 'HOST' if nothing is entered. | |
| **7.** | **CHECK THE CHANNEL SECURITY DATA (CSECD) RECORDS** | |
| **a)** | For each Triton terminal that supports AKDS, read its record in the Channel Security Configuration window. Records are read according to the Channel ID field. The value of the Channel ID must equal the key locator value found in the terminal's BASE24-atm Terminal Definition File (TDF) record, screen 9, Master Key field. | |
| **b)** | If a Channel Security Data record does not exist for this terminal, add it. | |
| **c)** | The Key Length of the Exchange Key should be set to **Double**.<br>The Key Length of the Inbound Key should be set to **Double**.<br>The MAC option must be set to **Pseudo Triple DES Macing with 32 bits**. TSS 6.2 and above supports this for Thales and Eracom, Atalla has had this support all along.<br>Modify these fields if necessary, save the record, and warmboot all IS processes. | |
| **8.** | **GENERATE AN RSA KEY PAIR FOR THE BASE24 SYSTEM** | |
| **a)** | To start the process of submitting a public key and getting it signed by Triton (via VeriSign), contact your Triton representative and request a Host ID. This will be used in the KEYGEN command and so must be obtained before you start the process.<br>Refer to the ***Applying for a Triton RKT Host Certificate*** document for detailed procedures on submitting your BASE24-eps system's public key and retrieving the appropriate certificates from Triton. To obtain a copy of this document, you must first execute a TDL (Triton Dynamic Language) license agreement with Triton. Contact your Triton representative for more information. | |
| **b)** | Generate an RSA public and private key pair for the BASE24 system. This is done one time only for a given BASE24 system unless the RSA key pair is corrupted in the TSS database.<br>To do so, go to the Process Control window of the ACI Desktop.<br>• Select **XMLS** in the Destination field.<br>• Select **Deliver** in the Command field.<br>• Select **Transaction Security** in the Component field.<br>• Enter **KEYGEN HOST, TRITON, HOSTID, URL, , 2048, , DATE>!** in the Command Text field. | ????? |

| NATIVE TRITON SETUP | | |
|---|---|---|
| | **Note: HOST** should be replaced with the value entered in the KEY SIGNER field on screen 22 of the ATD.<br><br>**Note: HOSTID** should be replaced by the Host ID value requested from Triton in step **8a)**.<br><br>**Note: URL** should be replaced with the url of the customer's organization, e.g. www.aciworldwide.com.<br><br>**Note: DATE** should be replaced with the effective date required for the public keys.<br><br>• Click on the green arrow in the upper left hand corner of the window to issue the command.<br><br>**Note:** The security device used to generate the RSA key pair is disabled during key generation, which may take up to 90 seconds. Ensure that the security device is PKI-enabled. | |
| **c)** | Access the Certificate Management window on the ACI Desktop to retrieve the BASE24 system public key certificate and format it in the manner expected by Triton's certificate authority, VeriSign. When the Certificate Management window is displayed, select Host name and select Triton as the device type in the Device Type field. Select the effective date required for the public key. The BASE24 system public key certificate is retrieved and displayed in the Root Host Public Key Data field. The key is displayed in PKCS#10 Certificate Request format, Base64 encoded. This is the message format required when sending the public key to Triton's certificate authority. | |
| **d)** | Copy the BASE24 system public key certificate in the Root Host Public Key Data field and paste it into a text file with a .csr file extension (e.g., <your-org-name>.csr). Submit your host public key certificate to Triton using Triton's secure web portal with VeriSign as described in Triton's ***Applying for a Triton RKT Host Certificate*** document. Make sure you remove any end-of-line characters as well as the "-----BEGIN NEW CERTIFICATE REQUEST-----" and "-----END NEW CERTIFICATE REQUEST-----" text from the certificate you submit. | |
| **e)** | When completing the CSR enrollment form, make sure you enter your assigned 16-digit host ID in the Host ID/Name field, your email address in the E-mail Address field, and the URL you used for the org-name variable when generating the host RSA key pair in the Company/Agency/Org field.<br><br>Triton will respond with an email that includes your signed host key certificate in text and in an attached cert.p7b file. Your assigned 16-digit host ID appears in the email salutation. If the Triton self-signed certificate is not included, you may need to send a separate request to Triton support to get it.<br><br>**Note**: Be sure to use the BASE64 certificate embedded in the body of the email rather than the attached .p7b file. The format in the attached .p7b file is incorrect. For the retrieval phase, you must reformat and copy the following certificates generated or returned by Triton and paste them in the appropriate fields on the Certificate Management window in the following steps. All certificates must be in PKCS #7 Cryptographic Message format before you paste them into BASE24. The following table shows you which field on the Certificate Management window to paste each certificate from Triton and in which of the following steps you paste the certificate.<br><br>**Note**: Before you can paste the bank host certificate and Triton CA certificate into BASE24, you must perform the steps listed below the table to reformat the certificate into | |

## NATIVE TRITON SETUP

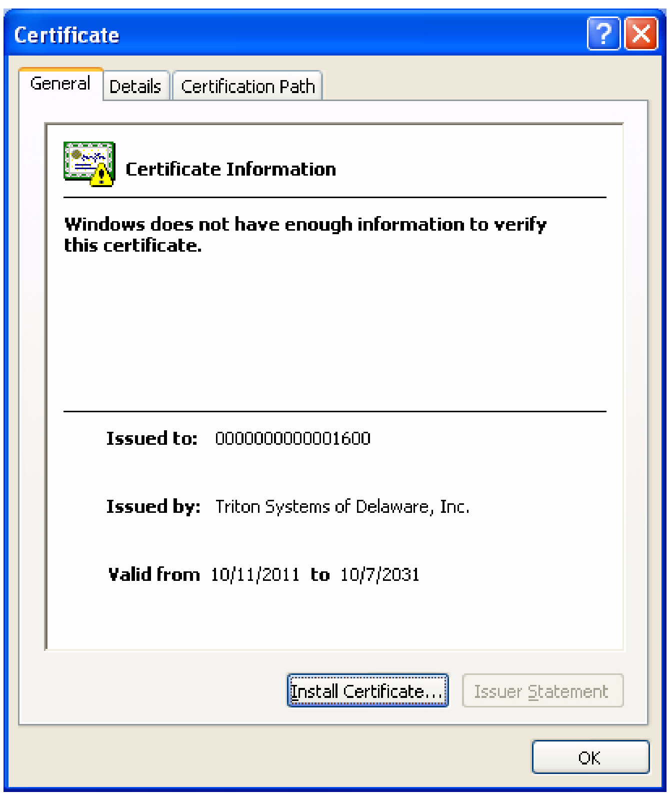
PKCS #7 Cryptographic Message format.

| Value (Triton) | Certificate Management Field |
|---|---|
| Triton self-signed CA certificate<br><br>Note: The self-signed certificate must be in text format, rather than p7b format, and contain signData. | Generate Message Digest > Root Certificate Authority Certificate (step **h**)<br><br>And/or<br><br>Validate Certificates/Signatures > Root Certificate Authority Certificate (step **k**) |
| BASE24 host certificate | Validate Certificates/Signatures > Host Primary Certificate (step **k)** |

**f)** Reformat the bank host certificate.

1) Save the bank host certificate from the email to a memorable name under Windows, such as “HOST_TRITON_ATM from converted 3-oct.cer”.

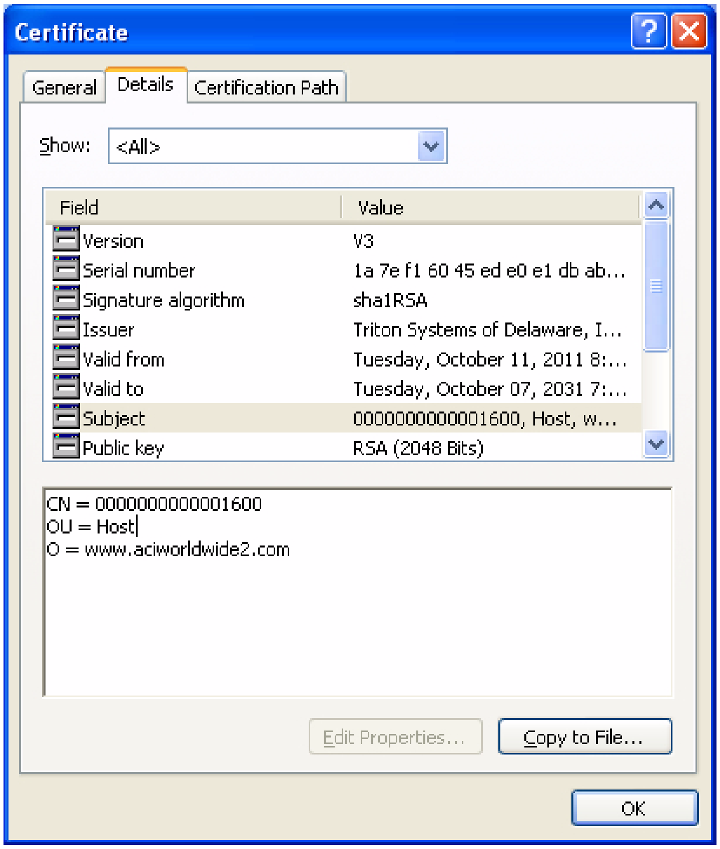
2) Double-click the file to start the Microsoft Management Console - Certificate view/ accept utility. Although this utility is normally used to install a certificate to be used for SSL communication, you are using it just to convert the certificate into a format compatible for use with BASE24.



# NATIVE TRITON SETUP

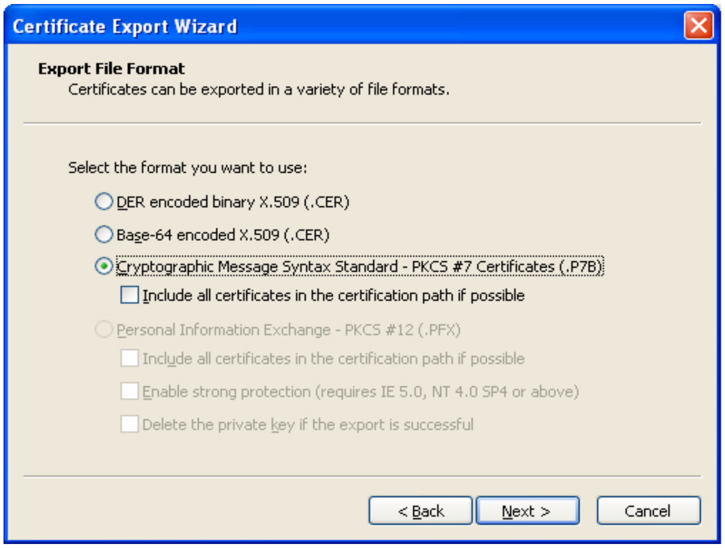
Note: ACI Worldwide makes no warranties about this third party utility.

3) Select Details and Subject to confirm the certificate is the one you expected to receive from Triton.

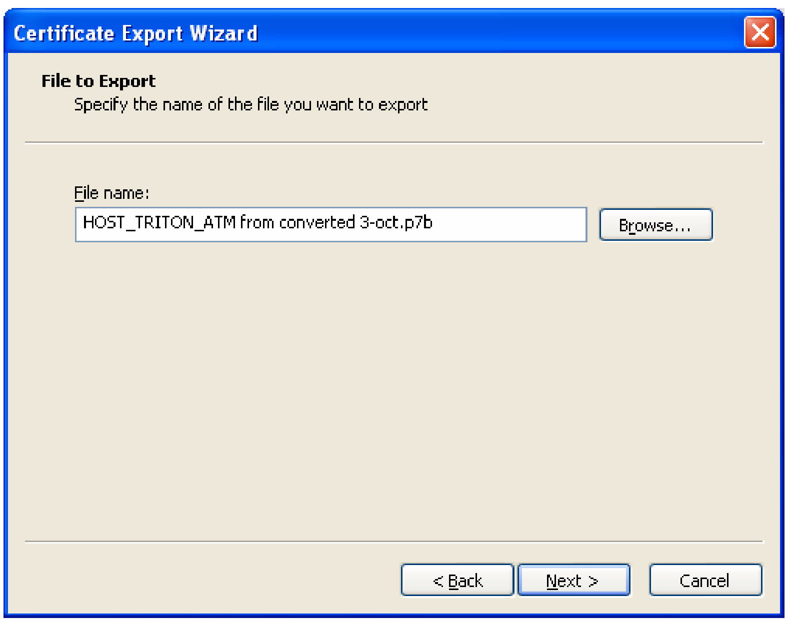



4) Select Copy to File to activate the export and conversion options. Select the "Cryptographic Message Syntax Standard - PKCS #7 Certificates (.P7B)" option and click Next.

## NATIVE TRITON SETUP




5) Specify the name of the file you want to export. Again, give it memorable name under Windows (e.g., HOST_TRITON_ATM from converted 3-oct.p7b) with a .p7b file extension and click Next.




6) The resulting file is saved in the P7B format, but is encoded in binary. You can then use a utility such as the ASN.1 Editor to convert the file from binary to PEM (character encoded) form, which is the format in which BASE24-eps expects to receive it.
The URL for downloading a free copy of ASN.1 is http://lipingshare.com/Asn1Editor/ for Asn1EditorSetup.msi.

Note: ACI Worldwide makes no warranties about this third party utility.

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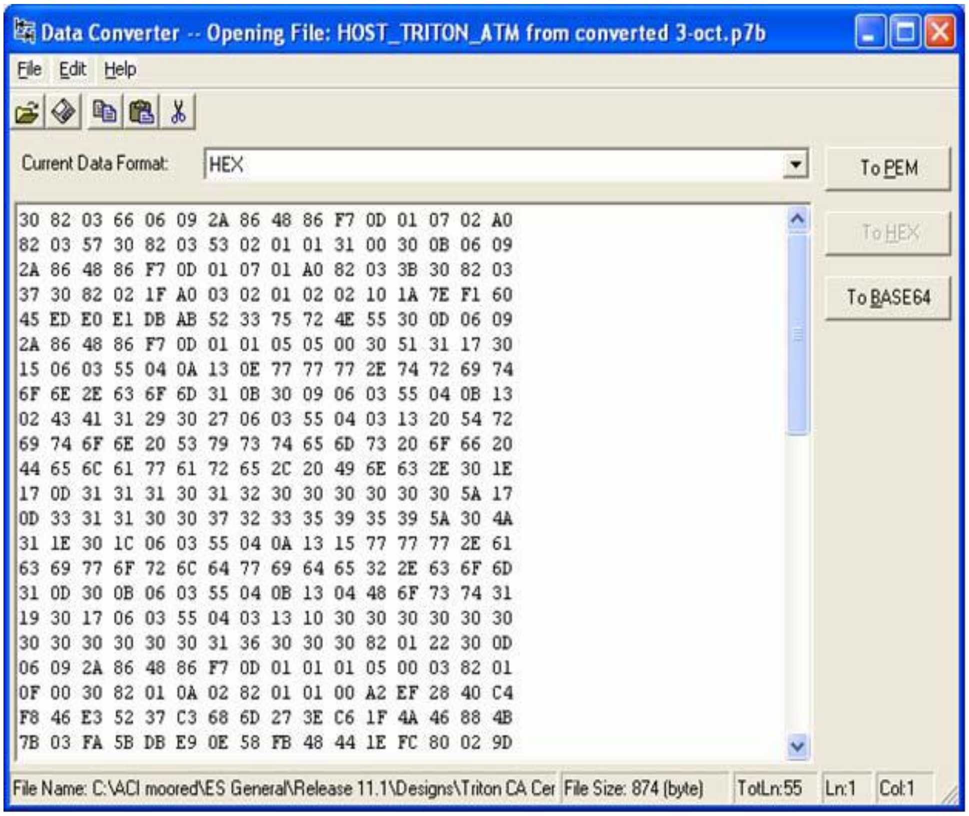
7) Once installed, you can use the ASN.1 Editor to confirm the certificate is correctly encoded. You can then use the DataConverter utility that comes with the ASN.1 Editor to change the format from binary to PEM format. BASE24 expects the certificate to be in PEM format (.pem).




8) Click on the To PEM button and then save the resulting file with a memorable name such as -- HOST_TRITON_ATM from converted 3-oct-PEM.p7b

9) This provides a result similar to the following. You must copy and paste the encoded part between the BEGIN and END markers into the Validate Certificates/ Signatures > Host Primary Certificate field on the Certificate Management window in step **k**.

-----BEGIN HOST_TRITON_ATM from converted 3oct.p7b-----
MIIDZgYJKoZIhvcNAQcCoiIDVzCCA1MCAQExADALBgkqhkiG9w0BBwGgggM7MIIDNzCC
Ah+gAwIBAgIQGo7xYEXt4OHbq1IzdXJOVTANBgkqhkiG1w0BAQUFADBRMRcwFQYDVQ
QKEw53d3cudHJpdG9uLmNvbTELMAkGA1UECxMCQ0ExKTAnBgNVBAMTIFRyaXRvbiBTe
XN0ZW1zIG9mIERlbGF3YXJlLCBJbmMuMB4XDTExMTAxMjAwMDAwMFoXDTMxMTAw
NzIzNTk1OVowSjEeMBwGA1UEChMVd3d3LmFjaXdvcmxkd2lkZTIuY29tMQ0wCwYDVQQL
EwRIb3N0MRkwFwYDVQQDExAwMDAwMDAwMDAwMDAxNjAwMIIBIjANBgkqhkiG9w0
BAQEFAAOCAQ8AMIIBCgKCAQEAou8oQMT4RuNSN8NobSc+xh9KRohLewP6W9vpDlj7SE
Qe/IACndASdDDKDROp0eNaPZEzcZ2cEk9XWkI2ZYx8uSLFURnLgzSw0OpxCrYX6ps+fNGH
lVq+GsYbnx8bNU97vQXjjEDEMCxhFa3ntkeo/OoZwRZKdM6DACCRmAXsK3c9XIOPiWL8as
3j4P3gVTTIV7d/mVoeki6HekasosFC1u2KrNYRr7E39npQV4Vd19cM8af8aDISnlzZ4YINhd8c+
CnWF7kP8cw1GRmkieYsQGupvJnornBwfnwV2HEoYkNM7bNeFO6N+toV9l2yt3FewY4lYz1z
WwkQSIPqRTxx7QIDAQABoxIwEDAOBgNVHQ8BAf8EBAMgB4AwDQYJKoZIhvcNAQEFB
QADggEBAJmdQKC5yCpghLxN5P4MKmDH/Dj9Asl7tVqvYOuR4JMtTM8Shanhg7ZgibtWyND

| NATIVE TRITON SETUP | | |
|---|---|---|
| | EJJg9cyTbvkp7NcArHmSKnmx1euQ3WJ5p7G095NDm4tsH85/di0Fh7gF9RnZ2S06Iyc2CTxQZS<br>PlQnv4KJgm19C+sHF/cpA7XjtkH1NokWr2+aYj3KSvumsL8kJccUMMkTWStqKF4b2c6fJZ9d2k<br>jVyiydtaCPgLRRQCbf9f8SAKn+7+6vIwrhFSZFHnVTsgCrc1MBrn+Pf34R4W2/YQOfc1VAXOJ<br>5QIYR5OzrIhFxrxTi0YaeQHQocT/cl/vew8HmkgTdorRogMUgPY8EdF7KaExAA==<br>-----END HOST_TRITON_ATM from converted 3oct.p7b----- | |
| **g)** | Reformat the Triton CA certificate.<br><br>1) Make sure you receive the Triton CA certificate in a suitable format, such as P7B.<br><br>2) Save the certificate to a file with a memorable name such as -- TritonCaCert.p7b<br><br>3) Use the ASN.1 associated tool DataConverter to change it from binary to PEM format (same as above for the bank host certificate), and save to another memorable file name (e.g., TritonCaCert-PEM.p7b).<br>This provides a result similar to the following. You must copy and paste the encoded part between the BEGIN and END markers into the Generate Message Digest > Root Certificate Authority Certificate field on the Certificate Management window in step **h** and/or into the Validate Certificates/Signatures > Root Certificate Authority Certificate on the Certificate Management window in step **k**.<br><br>-----BEGIN TritonCaCert.p7b-----<br>MIIDzwYJKoZIhvcNAQcCoIIDd62cA7wCAQExADALBgkqhkiG9w0BBwGgggOkMIIDoDCCA<br>oigAwIBAgIQKlyQxEZNoqDYdMPs9lOQ7zANBgkqhkiG9w0BAQUFADBRMRcwFQYDVQQ<br>KEw53d3cudHJpdG9uLmNvbTELMAkGA1UECxMCQ0ExKTAnBgNVBAMTIFRyaXRvbiBTeX<br>N0ZW1zIG9mIERlbGF3YXJlLCBJbmMuMB4XDTA3MTExMjAwMDAwMFoXDTMyMTExM<br>TIzNTk1OVowUTEYVBUGA1UEChMOd3d3LnRyaXRvbi5jb20xCzAJBgNVBAsTAkNBMSkwJ<br>wYDVQQDEyBUcml0b24gU3lzdGVtcyBvZiBEZWxhd2FyZSwgSW5jLjCCASIwDQYJKoZIhvc<br>NAQEBBQADggEPADCCAQoCggEBAMiJYkYaZDoYRy1qGXYh19Hq0buOxeYf1Npb6GdGz<br>OdvZfXHEASX4faZAv2UWXf1MuUI723jm8NkKV2g2W4/CmLJ0BNkytQrLP7yvsZPRktK+7u<br>MoMzyAXeN16aScEUsbwP/oOHtM9lIjZnsCwfbmaLkic9h7utipJUpIc5IPXDK8d7CR0DXL6alaC<br>Bqmvz4+RS2Xkg7qq+UhuAdlOiOMek6qVw8xP70yYmAP1BfspEja2ubjvlhgj2Jc9UjRgDU8v9Hp<br>R4clyuH7iHuVUwHmYXO8uDStZZqEA5aqcLlscRHEDB6X/bpJR48rfrvgYJIhLEBPrIG3jBIFtGz<br>bPWF/eECAwEAAaN0MHIwEgYDVR0TAQH/BAgwBgEB/wIBADAOBgNVHQ8BAf8EBAMC<br>AQYwLQYDVR0RBCYwJKQiMCAxHjAcBgNVBAMTFVByaXZhdGVMYWJlbDQtMjA0OC0<br>1NjAdBgNVHQ4EFgQUo1T6IKdSPAgCr7GL/ECaWlUHBAswDQYJKoZIhvcNAQEFBQADggE<br>BABYYfoxRkdVPrqV++ffNH+tYsA6O7nhuA7U5OmKgMRNxsu8XmG+ZXHqT3E5hqt1Dh9qd<br>6r/uwoVOKb3oUabtNbpPBvXXyJSk7XH4rsp5a1KoWqq6yET8v9ZUv4+AqHjqIQ9iRSpVIdkBQ<br>rjTfVlnkO2p7zxkqnCsbxXHaeJBpDQF6otg3wK7NvbrgnuBWIzB1Wnaqf0lF9PR/6oXhquFymxa<br>Uwm8qhUN7n8sFSP7eo59TwYQk0gqrHmnYJjaS4Agp9Rq6p+A+6Pzb+Y5zN2a6Yh9d27MuOjk<br>Y8cOKSyT5uk5HVdJN+yESHclkGPoKZSOw46SoN6CJh9uAgCBFl/MxR8xAA==<br>-----END TritonCaCert.p7b----- | |
| **h)** | If you are using a Thales e-Security HSM, you can skip to step **k**.<br><br>If you are using an Atalla NSP, create a message digest of the root certificate authority certificate (Triton selfsigned CA Certificate).<br><br>On the Generate Message Digest tab of the Certificate Management window, perform the following:<br><br>1) Copy the root certificate authority certificate returned from Triton into the Root Certificate Authority Certificate field. You can click Expand View to expand this field if needed. | |

| NATIVE TRITON SETUP | | |
|---|---|---|
| | Note: If the cryptogram contains spaces between blocks of hexadecimal characters, the spaces are automatically removed in the message sent to the security device.<br>2) Click save.<br>The NSP uses command 126 to create a message digest value that is displayed in the Message Digest field on this tab. | |
| **i)** | Using the SCT, encrypt the value returned in the Message Digest field from the previous step under variant 30 of the MFK. Use the Utilities Menu--> Encrypt Challenge function on the SCT to encrypt the message digest value. You must input the message digest as two 16 byte values (Left Challenge and Right Challenge). Using the SCA instead, with the SCA connected, enter a card and PIN. Select security management. Select Calculate AKB/Cryptogram. For Cryptogram Type select Challenge/Response and select Next. Enter the leftmost 16 characters and select Next. When the next screen is displayed select OK and put in a second Security Administrator card and enter its PIN. Select Confirm. The response and check digits will be displayed. Select Save to save the response in the Data Table. Repeat the process for the 16 rightmost characters. The results are two 16 byte values that comprise the root certificate hash value. | ??? |
| **j)** | Copy or enter the 32 byte (16 byte left and 16 byte right) values from the previous step into the Root Certificate Hash Value field on the Validate Certificates/Signatures tab. | |
| **k)** | On the Validate Certificates/Signatures tab of the Certificate Management window perform the following:<br>1) Copy the root certificate authority certificate returned from Triton in step **e** into the Root Certificate Authority Certificate field. You can click Expand View to expand this field if needed.<br>2) Copy the BASE24 system host certificate returned from Triton in step **e** into the Host Primary Certificate field. You can click Expand View to expand this field if needed.<br>3) Click save.<br>When you click Save, the following processing is performed:<br>a) The root certificate authority certificate is validated using the hash value input from step 9 (for Atalla NSPs only), the self-signed signature of the certificate is validated, and the certificate authority public key is stored in the Host Public Key Security File (HOST_PUBLIC_KEY).<br>b) The signature of the BASE24 system (host) primary certificate is validated using the primary certificate authority's public signature verification keys stored in step a. The root BASE24 system primary public key signature is stored in the HOST_PUBLIC_KEY.<br>The validity date range for the certificate is parsed and stored in the HOST_PUBLIC_KEY as the begin and end dates for the certificate. These are the "notBefore" and "notAfter" dates in the PKCS #7 certificate format.<br>30 1E validity<br>17 0D notBefore August 13, 2001, 11:50:05am GMT<br>30 31 30 38 31 33 31 35 35 30 30 35 5A<br>17 0D notAfter August 13, 2011, 11:50:05am GMT<br>31 31 30 38 31 33 31 35 35 30 30 35 5A | |

| NATIVE TRITON SETUP | | |
|---|---|---|
| | Note that the certificate for the certificate authority is parsed and only the public key value is stored in the HOST_PUBLIC_KEY. For the root BASE24 system (host) primary public key, the full certificate signature is stored in the HOST_PUBLIC_KEY. | |
| **l)** | Once the above steps are completed, the following key data is entered into the database for the AKDS process or in the tamper-resistant memory of the Hardware security module:<br>• Certificate authority public key<br>• Root BASE24 system’s public key signature (signed by the certificate authority’s private key)<br>• The BASE24 system’s public signature verification key certificate beginning and ending validity dates<br>• Certificate authority’s public signature verification key<br>• The root BASE24 system’s private signature generation key | |
| **9.** | **DISABLE THE KEYGEN COMMAND (OPTIONAL)** | |
| | After you have successfully generated an RSA key pair for the BASE24 system and have had the public key signed by the certificate authority, you may optionally disable use of the KEYGEN command.  This helps prevent the unintentional generation of a new RSA key pair, which, if it occurred, would require you to repeat the public key signing process. The KEYGEN command can be re-enabled if it is needed at a later time. | |
| **d)** | Go to the Disable KEYGEN window of the ACI Desktop. | |
| **e)** | Enter the value entered in the KEY SIGNER field on screen 22 of the ATD in the Host Name field<br><br>Select **Triton** in the Device Type field.<br><br>Select **KEYGEN command is disabled** in the KEYGEN Command Status field. | |
| **f)** | Save the record. | |
| **End of Section** | | |

| TRITON TERMINAL DOWNLOADS | | |
|---|---|---|
| **1.** | **DOWNLOAD ALL KEYS** | |
| | You may load all of the keys (public, master, MAC and communication keys) using a single NCPCOM command, the LOAD ALL KEYS command. | |
| **a)** | Go to NCPCOM | |
| **b)** | Enter DELIVER PROCESS <process-name>, “9500LOADALLKEYS <terminal-id>” | |
| **c)** | After the next transaction is received by the device handler, an AKDS load will be initiated. | |
| **d)** | Check the event log for the successful completion of the download. | |
| **End of Section** | | |

| MODIFICATION HISTORY | | | |
|---|---|---|---|
| **REV DATE** | **INITIALS** | **VERSION** | **HISTORY** |
| 03FEB2005 | RDS | 1 | Initial release. |
| 12JAN2006 | RDS | 2 | Added ATM DEVICE SETUP section.<br>Removed SET PROCESS SERVICE IS and SET PROCESS SERVICE AKDSIS lines from step 2.b) of AKDS PROCESS SETUP. Replaced with SET PROCESS SERVICE IS:AKDSIS.<br>Added reference to $<VOL>.BA60UD08.SCNTSSRR to step 1.a) of DIEBOLD 10XX/478X SETUP and step 1.a) of NCR 5XXX SETUP.<br>Noted that the SCT automatically assumes variant 30 of the MFK in step 8.f) of DIEBOLD 10XX/478X SETUP and step 8.g) of NCR 5XXX SETUP.<br>Removed extraneous comma from Diebold KEYGEN command. |
| 03MAY2006 | JEW | 3 | Add reference to link processes being required between nodes to step 2.c) of AKDS PROCESS SETUP. |
| 08AUG2006 | JEW | 4 | Reworded section on applying TSS fixes. |
| 12OCT2006 | JEW | 5 | Added a manual step required for applying TSS fixes to the AKDS Process Setup section. |
| 20NOV2006 | JEW | 6 | Added details of the Shared Signatures, Shared Certificates and the NCR 5XXX Certificates AKDS solutions. |
| 25MAR2010 | KKE | 7 | Corrected DIEBOLD host public key generation instructions to remove an erroneous exclamation point. |
| 18APR2012 | JEW | 8 | Added Triton, Tidel, Diebold Signatures, Wincor support of NCR ATMS and NCR Enhanced AKDS solutions. Updated Key Signer ATD field details. |

## End of Section

## End of Document